# Siting in the Dark — first-pass feasibility diagnostics

Can public data support the news-supply × data-center-siting question?

3 November 2026

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**Scope.** This memo reports feasibility diagnostics only. All results are correlational: "is associated with" throughout. Nothing here estimates a causal effect.

**Status of this version.** Grid-infrastructure controls are included: county substations ≥230 kV, km of ≥345 kV lines, and project distance to the nearest ≥230 kV substation. Transmission lines come from Esri's archived copy of HIFLD (last updated 2024-09-30); substations come from OpenStreetMap (pulled 2026-11-03), because the HIFLD substation layer is no longer public. About 22% of OSM substations (16,502 of 73,404) have no voltage tag, so

≥230 kV counts are a lower bound [output/logs/30_pull_grid_infra.txt]. Project distances above 500 km (AK/HI campuses) are set to missing. Adding grid left every conclusion unchanged relative to the earlier no-grid build; the coefficients that moved are noted where they appear.

# 1. Bottom line

**Does the project have legs on these data?** *Only if* project-level exposure and outcome measures are added. On public county-level news-supply counts plus Aterio alone, the answer is **no** for the closure design and close to no for the cross-sectional sorting design.

- **Design A (newspaper closures as shocks) is not powered.**
  - **Closure events by definition.**

| Event | Campuses | Counties | Failed campuses among them |
|---|---|---|---|
| Any newspaper-establishment decline within ±3 years (QCEW) | 635 of 1,066 dated (60%) | 240 | — |
| 1→0 transition (county loses its last newspaper) within ±3 years | 61 | 28 | 7 |

    [a3_overlap_totals, a3_outcome_cells]
  - **Why the large numbers mislead.** The 635 figure is mostly routine attrition in the large counties where campuses sit, not desert formation. Nationally, the median decline event happens in a county with 3 establishments the year before, and the 90th percentile in one with 16 [a2_decline_event_prior_level].
  - **The 1→0 event study.** It has only 23 treated county-years with any announcement, so it was not estimated [43_extensive.txt].
- **The single biggest obstacle is measurement on both sides of the regression.**
  - **The news measure.** Public newspaper counts are establishment counts, not newsroom capacity or coverage. Census CBP has censored the cells that matter since 2017: county cells with <3 establishments are not published.
  - **The outcome.** Aterio's outcomes are snapshot stage codes, not local decisions. Dated projects are overwhelmingly 2023–26 (951 of 1,066 dated campuses announced 2010+, 89%), and only 87 dated campuses have failed.
- **What the diagnostics show.**
  - **News deserts differ sharply from other counties.** Zero-newspaper counties have about a tenth the population of 2+ counties (mean 23,487 vs 250,731), are far more rural (mean RUCC 6.2 vs 3.8), and have land worth less than half as much [b3_balance_all_counties].
  - **Raw associations do not survive comparison within states.** Associations between news supply and announcements are strong in raw data, mostly vanish with state×year fixed effects, and behave like the warehousing placebo.
  - **The one robust within-county association runs the wrong way for the hypothesis.** It is in the any-announcement LPM with county + state×year FE, and survives dropping VA and/or TX. *More* newspapers per capita, and *not* being a zero-newspaper county, go with *more* announcements.
  - **Failure.** It is not associated with news supply in any specification.
  - **Secrecy (P3).** Undisclosed developers are, if anything, slightly *more* common where news is thin, and the association is not robust.

# 2. Data built

## 2.1 Sources pulled

| Source | What | Coverage / vintage | Known problems |
|---|---|---|---|
| Aterio inventory (licensed) | 7,544 facilities (7,132 US) → 2,425 US campuses | snapshot 2026-06-30; second snapshot 2026-02-09 (12 fields) | 30% of facilities lack announced date; record-creation dates start 2024-07-01 |
| **BLS QCEW** (primary news measure) | private establishments, NAICS 511110 (to 2021) / 513110 (2022+), county × year | 2005–2025 | 2021→22 series break (513110 adds internet newspaper publishers); employment suppressed in 85–90% of county cells |

| Source | What | Coverage / vintage | Known problems |
|---|---|---|---|
| Census CBP | NAICS 511110 county establishments/employment | 2005–2023 | **from 2017 cells with <3 establishments are not published**; still coded 511110 in 2022–23 |
| Census PEP | county population | 2005–2025 | CT planning regions from 2020s vintages (crosswalked) |
| ACS 2017–21 5-yr | population, income, age, education, race | cross-section | bulk summary file (Census API now needs a key) |
| USDA ERS | RUCC 2023, UIC 2024 (RUCC/UIC 2013 for CT legacy counties) | cross-section | |
| Presidential returns | R share 2016/2020/2024 | MEDSL GitHub 2000–16; unofficial compilation for 2020/2024 | official MEDSL Dataverse file needs a guestbook form; AK only statewide |
| Census of Governments 2022 | general-purpose governments per county | cross-section (CT from 2017) | national counts match published totals exactly |
| USDA NASS (bulk) | ag land value $/acre, Census of Agriculture 2012/17/22 | 3,071 of 3,143 counties | VA independent cities, AK missing |
| EIA-861 | utility C&I prices; county via service territory | 2019–2025 | county-name match 99.9%; Aterio utility-code match 90.4% |
| WRI Aqueduct 4.0 | baseline water stress (project + county) | 2023 release | |
| FCC CDBS | full-power TV/AM/FM by community of license | frozen 2024-01-09 | LMS (current) blocked to scripts |
| FCC Form 477 | broadband (collected; not used in regressions) | June 2021 | National Broadband Map blocked to scripts |
| LoC Directory of US Newspapers | titles with start/end years | partial: 2000+ national subset + all TX/VA | end years rarely updated (below) |
| Data Center Watch | 16-project list from free report | 2023–Q1 2025 | later lists are paid |
| Medill State of Local News 2025 | national totals only | report PDF | no state/county table published |
| Grid (HIFLD lines via Esri archive + OSM substations) | county ≥230 kV substations, km ≥345 kV lines, project distance to ≥230 kV substation | lines 2024-09-30; OSM 2026-11-03 | HIFLD substations not public anymore; 22% of OSM substations lack voltage (≥230 kV counts are a lower bound) |

All raw files are saved untouched under `data/raw/<source>/`, each with a `SOURCE.md`. Data live in Dropbox (`~/Library/CloudStorage/Dropbox/data-centers-newspaper/data`).

## 2.2 Why QCEW, not CBP, is the primary news measure

The brief planned CBP as the backbone. The CBP files themselves show it cannot play that role after 2016 [`cbp_published_cells_by_year`]:

| year | county_cells | total_est | min_est | n_lt3 |
|---|---|---|---|---|
| 2005 | 2707 | 8395 | 1 | 1886 |
| 2006 | 2714 | 8547 | 1 | 1899 |
| 2007 | 2707 | 8659 | 1 | 1855 |
| 2008 | 2679 | 8375 | 1 | 1890 |
| 2009 | 2659 | 8041 | 1 | 1875 |
| 2010 | 2655 | 7794 | 1 | 1881 |
| 2011 | 2653 | 7671 | 1 | 1908 |
| 2012 | 2657 | 7624 | 1 | 1906 |
| 2013 | 2644 | 7479 | 1 | 1910 |
| 2014 | 2633 | 7384 | 1 | 1919 |
| 2015 | 2606 | 7164 | 1 | 1912 |
| 2016 | 2588 | 7495 | 1 | 1889 |
| 2017 | 669 | 4720 | 3 | 0 |
| 2018 | 626 | 4159 | 3 | 0 |
| 2019 | 645 | 4334 | 3 | 0 |
| 2020 | 619 | 4153 | 3 | 0 |
| 2021 | 580 | 3803 | 3 | 0 |
| 2022 | 574 | 3681 | 3 | 0 |
| 2023 | 561 | 3555 | 3 | 0 |

- **What breaks.** Through 2016 the minimum published establishment count is 1. From 2017 it is 3, and county cells with a 511110 row fall from 2,588 to 669. A county without a row in 2017+ may have 0, 1 or 2 newspaper establishments, so the CBP zero flag, 1→0 transitions and per-capita counts are unusable after 2016.
- **What QCEW provides instead.** It publishes the establishment count for every county-industry cell, suppressing only employment and wages. It also runs to 2025, which covers the 2024–25 announcement surge.
- **How the two compare.** In 2005–2016 they correlate at 0.95 across county-years. QCEW finds slightly fewer zero counties because it counts UI-covered worksites [`news_qcew_vs_cbp_agreement`].
- **How CBP is kept.** It is retained as `c_*` variables: exact counts through 2016, and a "≥3 establishments" indicator that is consistent across all years.

| year | est_total | zero | one | ge3 | wh_est |
|---|---|---|---|---|---|
| 2005.000 | 9182.000 | 347.000 | 1202.000 | 911.000 | 13378.000 |
| 2006.000 | 9229.000 | 348.000 | 1224.000 | 906.000 | 13815.000 |
| 2007.000 | 9103.000 | 361.000 | 1223.000 | 899.000 | 14362.000 |
| 2008.000 | 8995.000 | 359.000 | 1246.000 | 883.000 | 14932.000 |
| 2009.000 | 8639.000 | 376.000 | 1267.000 | 856.000 | 14868.000 |
| 2010.000 | 8338.000 | 382.000 | 1294.000 | 836.000 | 14897.000 |
| 2011.000 | 8115.000 | 399.000 | 1295.000 | 810.000 | 14955.000 |
| 2012.000 | 8020.000 | 418.000 | 1296.000 | 796.000 | 15286.000 |
| 2013.000 | 7847.000 | 432.000 | 1290.000 | 784.000 | 15462.000 |
| 2014.000 | 7673.000 | 456.000 | 1296.000 | 748.000 | 15708.000 |
| 2015.000 | 7522.000 | 466.000 | 1311.000 | 736.000 | 16143.000 |
| 2016.000 | 7408.000 | 483.000 | 1318.000 | 722.000 | 16557.000 |
| 2017.000 | 7183.000 | 504.000 | 1318.000 | 682.000 | 16897.000 |
| 2018.000 | 6916.000 | 530.000 | 1333.000 | 639.000 | 17613.000 |
| 2019.000 | 6705.000 | 545.000 | 1359.000 | 608.000 | 18150.000 |
| 2020.000 | 6556.000 | 582.000 | 1363.000 | 588.000 | 18606.000 |
| 2021.000 | 6398.000 | 614.000 | 1345.000 | 558.000 | 19325.000 |
| 2022.000 | 6858.000 | 629.000 | 1317.000 | 587.000 | 20457.000 |
| 2023.000 | 6707.000 | 686.000 | 1283.000 | 563.000 | 21036.000 |
| 2024.000 | 6498.000 | 753.000 | 1260.000 | 532.000 | 21517.000 |
| 2025.000 | 6255.000 | 816.000 | 1267.000 | 521.000 | 21989.000 |

## 2.3 Geography

Everything is on the **2020 county universe**: 3,143 counties for the 50 states + DC, with Connecticut's 8 legacy counties [`code/geo.py`].

- **Renames.** 02270→02158, 46113→46102, 51515→51019 and 02201→02198 are mapped one-to-one.
- **Splits.** Valdez-Cordova, Skagway-Hoonah-Angoon and Wrangell-Petersburg are allocated by 2020 population.
- **Connecticut planning regions.** These are used by PEP 2020s vintages, CBP 2022–23, QCEW 2024–25, RUCC 2023, CoG 2022 and ACS 2022+. They are allocated to legacy counties with Census's town-level crosswalk and 2020 town populations; every town nests in one region and one legacy county.
  - CBP for CT in 2022–23 is set to missing, because the <3 censoring is defined at region level.
  - QCEW CT changes for 2023→24 are set to missing.
  - Sources that ship both CT geographies keep the legacy rows only.

## 2.4 Aterio: unit of analysis and merges

**Campus rule** [`code/05_build_aterio_campus.py`]:

- **Campus ID.** The campus is `ATERIO_DATA_CENTER_CAMPUS_UID`; the 1,425 US facilities with no campus UID are their own campus.
- **Location and dates.** Announcement date is the earliest building's. County, coordinates, utility and BA are those of the earliest-announced building.
- **Size and flags.** MW is the sum of `SELECTED_POWER_CAPACITY_MW`. AI and BTM equal 1 if any building is flagged.
- **Failure.** `failed` = all buildings Not Approved/Withdrawn or Cancelled. The "any building" version differs for 1 campus.

| metric | value |
|---|---|
| US facilities | 7132.000 |
| US campuses | 2425.000 |
| campuses with 1 building | 1524.000 |
| campuses with 2-5 | 577.000 |
| campuses with 6+ | 324.000 |
| max buildings in a campus | 60.000 |
| mean buildings per campus | 2.941 |

**Stage values are not those listed in the brief.** The file has Announcement, Active, Construction, Not Approved/Withdrawn, Delayed, Land Bank and Cancelled; there is no "Announcement of Development".

| DATA_CENTER_STAGE | CA | US | total |
|---|---|---|---|
| Announcement | 200 | 3631 | 3831 |
| Active | 194 | 2026 | 2220 |
| Construction | 14 | 766 | 780 |
| Not Approved/Withdrawn | 2 | 552 | 554 |
| Delayed | 2 | 84 | 86 |
| Land Bank | 0 | 54 | 54 |
| Cancelled | 0 | 19 | 19 |
| Total | 412 | 7132 | 7544 |

**County assignment.**

- **Method.** Point-in-polygon on Aterio's lat/lon assigns 7,129 of 7,132 US facilities; one more uses Aterio's own FIPS, and 2 have no location.
- **Agreement with Aterio's FIPS.** The two agree for 93.7% of facilities (443 disagree), and 95% of disagreements are adjacent counties.
- **Spot checks cut both ways.** In Storey County, NV (Tahoe-Reno Industrial Center) the coordinates are right. For a "Casper, WY" record the coordinates sit in Cheyenne.
- **Choice.** Point-in-polygon is primary, and Aterio's FIPS is carried as `fips_aterio` (68 dated campuses in the D sample differ) [`05_build_aterio_campus.txt`, `44_intensive.txt`].

**Merge diagnostics.**

| Step | In | Matched | Dropped / note |
|---|---|---|---|
| US facilities → campuses | 7,132 | 2,425 campuses | — |
| campuses with county | 2,425 | 2,423 | 2 no coordinates/FIPS |
| campuses with announced date | 2,425 | 1,068 | 1,357 undated (mostly Active) |
| dated campuses with QCEW news in announcement year | 1,068 | 1,065 | 2 no county, 1 announced 2002 (pre-2005) |
| 2026 announcements | 227 | use 2025 news | flagged `news_year_lagged` |
| D sample after county controls | 1,065 | 1,049 | 16 missing controls |
| Feb-2026 snapshot UIDs | 5,870 | 5,767 in June | 103 dropped by June; 1,365 June facilities new since Feb |
| EIA utility code → EIA-861 | facilities with code | 90.4% | [`progress.md`, 31_pull_eia861] |
| Data Center Watch 16 projects → Aterio | 16 | 10 | 6 unmatched (name/place) |

## 2.5 Failed or partial sources

From `logs/data_failures.md`:

| Source | Problem | What was done |
|---|---|---|
| Census API | keyless requests now refused | bulk files (CBP, ACS SF, PEP) |
| CBP 2017+ | <3-establishment cells not published | QCEW primary |
| CBP ZBP (place level) | 2023 detail has only 243 ZIP×511110 cells, all ≥3 | **place-level news supply not constructible** |
| USDA NASS API | needs key | keyless bulk `qs.economics` file |
| MEDSL Dataverse | guestbook form required | MEDSL GitHub 2000–16 + unofficial 2020/24 compilation |
| MEDSL 2024 precincts | state totals did not reconcile | used for CT only |
| FCC LMS / Broadband Map | 403 to scripts | CDBS (2024-01) / Form 477 (2021) |
| LoC directory | full pull ≈4 h; one national page failed after retries | 2000+ national subset (may miss up to one page of records) + TX + VA |
| GDELT (BigQuery) | `bq` not installed | not attempted |
| Data Center Watch | no free list after Q1 2025 | 16-project report list |
| HIFLD substations | no public service | OSM Overpass substations; HIFLD lines from Esri archive |

# 3. Measurement findings

## A1. Validity against Medill

| measure | value | source / note |
|---|---|---|
| Medill 2025 report: counties with zero local news sources | 212 | report text p.6/p.16 (press release: 213) |
| Medill 2025 report: counties with one local news source | 1525 | report text p.16 (press release: 1,524) |
| Medill 2025: share of news deserts predominantly rural | almost 80% | report p.17 |

| measure | value | source / note |
|---|---|---|
| QCEW 2025: counties with 0 private newspaper establishments | 816 | this study |
| QCEW 2025: counties with exactly 1 | 1267 | this study |
| QCEW 2025: share of zero counties that are nonmetro (RUCC 4-9) | 0.723 | this study |
| QCEW 2025: zero counties with >=1 full-power TV station licensed | 58 | FCC CDBS |
| QCEW 2025: zero counties with >=1 full-power radio station | 609 | FCC CDBS |
| CBP 2016 (last uncensored): counties with 0 newspaper establishments | 554 | this study |
| CBP 2016: share of zero counties nonmetro | 0.758 | this study |
| QCEW 2016: counties with 0 | 484 | this study |
| CBP 2023: counties with no published 511110 cell (0, 1 or 2 establishments) | 2580 | CBP censors <3 |
| CBP 2023: counties with >=3 establishments | 555 | this study |

**QCEW flags 816 zero-establishment counties in 2025, nearly four times Medill's 212 news deserts.** It counts 1,267 counties with exactly one establishment, against Medill's 1,525 one-source counties.

**Where the two concepts diverge:**

- **Unit.** A QCEW/CBP "establishment" is a UI-reporting worksite classified as a newspaper publisher. A chain paper printed, edited and reported from a neighbouring county's office is an establishment there, not in the county it covers. Medill counts *titles and outlets serving* a county, including weeklies run by tiny or non-employer operations that never appear in establishment data.
- **Scope.** Medill also counts public broadcasters, digital sites and ethnic media; 513110 contains some digital-only newspapers only from 2022.
- **Coverage.** Establishments with no employees (sole proprietors) are not in QCEW or CBP at all.
- **Rurality.** The direction of the gap is as expected: the establishment measure over-calls deserts in small rural counties. 72% of QCEW zero counties are nonmetro, versus Medill's "almost 80%" of news deserts being predominantly rural.
- **Broadcast.** Of the 816 QCEW-zero counties, 58 have a full-power TV station licensed to a community there and 609 have a full-power radio station [FCC CDBS].
- **State comparison.** It is not possible: Medill publishes no state table, and its maps are images. QCEW zero counties by state are in [`a1_qcew_zero_by_state_2025`]; TX (97), GA (75) and VA (64) lead.

## A2. Closure events

| year | q_decl | q_1to0 | q_0to1 | c_decl | c_1to0 |
|---|---|---|---|---|---|
| 2006 | 285 | 17 | 16 | 295 | 32 |
| 2007 | 298 | 21 | 8 | 343 | 48 |
| 2008 | 289 | 19 | 21 | 486 | 57 |
| 2009 | 358 | 28 | 12 | 366 | 46 |
| 2010 | 294 | 18 | 12 | 298 | 27 |
| 2011 | 290 | 28 | 11 | 251 | 20 |
| 2012 | 267 | 33 | 13 | 368 | 39 |
| 2013 | 273 | 20 | 6 | 271 | 25 |
| 2014 | 273 | 32 | 8 | 258 | 30 |
| 2015 | 261 | 29 | 20 | 308 | 39 |
| 2016 | 259 | 31 | 13 | 246 | 39 |
| 2017 | 304 | 38 | 17 | nan | nan |
| 2018 | 336 | 45 | 19 | nan | nan |
| 2019 | 297 | 32 | 18 | nan | nan |
| 2020 | 279 | 50 | 13 | nan | nan |
| 2021 | 295 | 53 | 21 | nan | nan |
| 2022 | 223 | 54 | 39 | nan | nan |
| 2023 | 322 | 81 | 24 | nan | nan |
| 2024 | 364 | 95 | 27 | nan | nan |
| 2025 | 360 | 82 | 19 | nan | nan |

| rural_grp | counties | q_decl | q_1to0 | c_decl | c_1to0 | q_decl_per_county | q_1to0_per_county |
|---|---|---|---|---|---|---|---|
| metro (1-3) | 1186 | 3649.000 | 258.000 | 2036.000 | 95.000 | 3.077 | 0.218 |
| nonmetro adj/urban (4-7) | 902 | 1343.000 | 188.000 | 828.000 | 72.000 | 1.489 | 0.208 |
| rural (8-9) | 1055 | 935.000 | 360.000 | 626.000 | 235.000 | 0.886 | 0.341 |

- **Declines are common.** QCEW shows 223–364 county-year declines per year.
- **Full exits are rarer and rising.** 1→0 transitions run 17–95 a year and have risen sharply since 2020: 81 in 2023, 95 in 2024, 82 in 2025.

- **Rurality.** Declines concentrate in metros (3.1 per county over 2006–25 vs 0.9 in RUCC 8–9), but 1→0 exits are most frequent in the most rural counties (0.34 per county).
- **The 2022 dip** in declines (223) coincides with the NAICS change, which added establishments.

## A3. Overlap and power (Design A)

| spec | campuses_with_event | share | campuses_with_any_observable_year | distinct_counties_with_event | n_campuses |
|---|---|---|---|---|---|
| q_decl_w-3_3 | 635 | 0.596 | 1065 | 240 | 1066 |
| q_decl_w-3_0 | 602 | 0.565 | 1065 | 233 | 1066 |
| q_decl_w-3_3_no2022 | 610 | 0.572 | 1065 | 223 | 1066 |
| q_1to0_w-3_3 | 61 | 0.057 | 1065 | 28 | 1066 |
| q_1to0_w-3_0 | 54 | 0.051 | 1065 | 26 | 1066 |
| q_1to0_w-3_3_no2022 | 60 | 0.056 | 1065 | 27 | 1066 |
| c_decl_w-3_3 | 19 | 0.018 | 36 | 18 | 1066 |
| c_1to0_w-3_3 | 1 | 0.001 | 36 | 1 | 1066 |

| ann_year | campuses | q_decl_w-3_3 | q_1to0_w-3_3 | q_decl_w-3_0 | q_1to0_w-3_0 | failed | failed_and_q_decl_w-3_3 |
|---|---|---|---|---|---|---|---|
| 2002 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2009 | 1 | 1 | 0 | 1 | 0 | 0 | 0 |
| 2010 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2011 | 3 | 3 | 0 | 3 | 0 | 0 | 0 |
| 2013 | 1 | 1 | 0 | 1 | 0 | 0 | 0 |
| 2015 | 2 | 1 | 0 | 1 | 0 | 0 | 0 |
| 2016 | 5 | 2 | 1 | 2 | 1 | 0 | 0 |
| 2017 | 8 | 5 | 0 | 5 | 0 | 0 | 0 |
| 2018 | 7 | 6 | 0 | 5 | 0 | 0 | 0 |
| 2019 | 8 | 7 | 0 | 7 | 0 | 0 | 0 |
| 2020 | 10 | 8 | 0 | 6 | 0 | 0 | 0 |
| 2021 | 26 | 22 | 1 | 21 | 0 | 4 | 4 |
| 2022 | 44 | 38 | 4 | 29 | 0 | 1 | 1 |
| 2023 | 85 | 61 | 9 | 50 | 8 | 3 | 3 |
| 2024 | 230 | 156 | 18 | 147 | 17 | 16 | 11 |
| 2025 | 409 | 211 | 18 | 211 | 18 | 51 | 30 |
| 2026 | 225 | 113 | 10 | 113 | 10 | 12 | 6 |

| stage_grp | campuses | q_decl_w-3_3 | q_1to0_w-3_3 | q_decl_w-3_0 | q_1to0_w-3_0 |
|---|---|---|---|---|---|
| Active | 162 | 120 | 7 | 104 | 5 |
| Announcement | 584 | 333 | 32 | 327 | 29 |
| Construction | 199 | 110 | 11 | 103 | 10 |
| Delayed | 11 | 8 | 2 | 5 | 2 |
| Failed (withdrawn/cancelled) | 87 | 55 | 7 | 54 | 6 |
| Land Bank | 23 | 9 | 2 | 9 | 2 |
| Total | 1066 | 635 | 61 | 602 | 54 |

| cell | n |
|---|---|
| failed & decline(+/-3) | 55 |
| not failed & decline(+/-3) | 580 |
| failed & 1->0(+/-3) | 7 |
| not failed & 1->0(+/-3) | 54 |
| stalled_24 & decline(+/-3) | 52 |
| eligible for stalled_24 & decline(+/-3) | 227 |

- **Read the decline numbers as attrition, not desert shocks.** A ±3-year "decline" around a dated campus is the norm (60% of campuses), because campuses are in large counties where some establishment closes most years.
- **The desert-relevant event, 1→0, overlaps 61 campuses (5.7%) in 28 counties, of which 7 failed.**
- **Window truncation.** For 2024–26 announcements the window is right-truncated (QCEW ends 2025). The pre-only window (–3..0) gives 54 campuses.
- **CBP cannot contribute.** With exact counts only through 2016, it overlaps 36 dated campuses in total.
- **Conclusion.** A closures-as-shocks design with failure as the outcome would rest on about 7 treated failures. **It is not feasible with these data.**

## A4. Detection bias in Aterio

- **What the record-creation date can measure.** `RECORD_CREATED_DATE` begins on 2024-07-01, so for earlier announcements the "detection lag" is mechanically time-until-database-inception (median 18 months). The informative sample is the 765 dated campuses announced on or after 2024-07-01. Their median lag is 0.26 months, and the 75th percentile is 1.8 months [`a4_detect_lag_describe`].

| term | coef | se | p | n | sample | spec | outcome |
|---|---|---|---|---|---|---|---|
| q_pc10k | 0.322 | 0.437 | 0.463 | 765 | announced >= 2024-07-01 | bivariate | detect_lag_m |
| q_pc10k | 0.297 | 0.414 | 0.474 | 765 | announced >= 2024-07-01 | controls | detect_lag_m |
| q_pc10k | -0.248 | 0.342 | 0.468 | 762 | announced >= 2024-07-01 | controls + state FE + ann-month | detect_lag_m |
| q_zero | 0.564 | 0.499 | 0.259 | 765 | announced >= 2024-07-01 | bivariate | detect_lag_m |
| q_zero | 0.075 | 0.452 | 0.868 | 765 | announced >= 2024-07-01 | controls | detect_lag_m |
| q_zero | 0.195 | 0.393 | 0.621 | 762 | announced >= 2024-07-01 | controls + state FE + ann-month | detect_lag_m |
| q_pc10k | -0.333 | 1.740 | 0.848 | 1065 | all dated | bivariate | detect_lag_m |
| q_pc10k | 2.338 | 1.831 | 0.202 | 1065 | all dated | controls | detect_lag_m |
| q_pc10k | -0.116 | 0.364 | 0.751 | 1029 | all dated | controls + state FE + ann-month | detect_lag_m |
| q_zero | -3.621 | 1.436 | 0.012 | 1065 | all dated | bivariate | detect_lag_m |
| q_zero | -3.461 | 1.434 | 0.016 | 1065 | all dated | controls | detect_lag_m |
| q_zero | 0.049 | 0.420 | 0.907 | 1029 | all dated | controls + state FE + ann-month | detect_lag_m |

- **Detection lag.** In the post-inception sample, lag is not associated with news supply. Coefficients are under ±0.6 months and insignificant with and without controls and state + month FE.
- **The full-sample q_zero coefficient (–3.5 months).** It is significant only without FE. It reflects the pre-inception mechanics (zero-news projects are newer), and it disappears with announcement-month FE.

| q_grp_2025 | campuses | share_no_ann_date | share_undisclosed | share_mw_missing | share_no_end_user | share_permit_url |
|---|---|---|---|---|---|---|
| 0 | 221 | 0.271 | 0.222 | 0.140 | 0.932 | 0.362 |
| 1 | 271 | 0.218 | 0.232 | 0.151 | 0.915 | 0.343 |
| 2+ | 1931 | 0.641 | 0.098 | 0.114 | 0.961 | 0.175 |

- **Missingness by news group.**
  - **Announced dates.** Campuses in 2+ counties are much *more* likely to lack an announced date (64% vs 27% in zero counties). This is because the undated stock is old, active metro data centers.
  - **Undisclosed providers and permit URLs.** Undisclosed providers are more common in zero/one counties (22–23% vs 10%), and permit URLs are more often present there (36% vs 18%).
  - **Interpretation.** These gaps are compositional (vintage, project type), not an obvious detection gradient.

| q_grp_2025 | dated_campuses | failed | failed_share |
|---|---|---|---|
| 0 | 161 | 17 | 0.106 |
| 1 | 212 | 13 | 0.061 |
| 2+ | 693 | 57 | 0.082 |

| q_pcq | facilities | late_added | share_late_added |
|---|---|---|---|
| 1.000 | 765.000 | 53.000 | 0.069 |
| 2.000 | 2298.000 | 242.000 | 0.105 |
| 3.000 | 686.000 | 118.000 | 0.172 |
| 4.000 | 287.000 | 43.000 | 0.150 |

- **Failed projects are not under-represented in low-news counties.** Among dated campuses the failed share is 10.6% in zero counties vs 8.2% in 2+ counties. Counted per 100 active campuses the gap is huge (41 vs 6), but that compares a recent flow to an old stock and is not informative [`a4_failed_vs_active_by_group`].
- **Late-added records.** Of the June-2026 facilities absent from the Feb-2026 snapshot, 456 were announced before 2026. Their share is *lowest* in the lowest-news quartile (6.9%) and highest in quartiles 3–4 (15–17%).
- **What this implies.**
  - **What it does not show.** There is no sign that Aterio detects projects later, or misses failures, in low-news counties, so detection does not obviously bias C–E toward finding a low-news association.
  - **What it cannot rule out.** Projects never detected at all are, by construction, unobservable here.
  - **Where Aterio finds projects.** Aterio's datasheet links show it relies heavily on company sites and government records, not only news: 25% of facilities cite at least one local-news URL, 37% a company site and 18% a government/agency page [`aterio_links_datasheet_type_shares`]. That plausibly explains the absence of a detection gradient.

## A5. Stage coding and outcome definitions

| stage | cancel_date=0 | withdraw_date=0 | cancel_date=0 | withdraw_date=1 | cancel_date=1 |
|---|---|---|---|---|---|
| Active | 2026 | 0 | 0 | 0 | 2026 |
| Announcement | 3631 | 0 | 0 | 0 | 3631 |
| Cancelled | 1 | 0 | 4 | 14 | 19 |
| Construction | 766 | 0 | 0 | 0 | 766 |
| Delayed | 84 | 0 | 0 | 0 | 84 |
| Land Bank | 54 | 0 | 0 | 0 | 54 |
| Not Approved/Withdrawn | 30 | 512 | 5 | 5 | 552 |
| All | 6592 | 512 | 9 | 19 | 7132 |

- **Withdrawn records.** Of 552 Not Approved/Withdrawn facilities, 512 have only a withdrawn date, 5 only a cancel date, 5 both, and **30 neither**.
- **Cancelled records.** Of the 19 Cancelled facilities, 1 has no cancel date.
- **Other stages.** No other stage carries cancel or withdraw dates, so the coding is internally consistent apart from these 31 date gaps. `failed` therefore uses the stage, not the dates.

| index | months |
|---|---|
| count | 955.000 |
| mean | 14.904 |
| std | 16.029 |
| min | -16.690 |
| 10% | 1.051 |
| 25% | 4.238 |
| 50% | 10.086 |
| 75% | 20.304 |
| 80% | 23.031 |
| 90% | 35.174 |
| max | 128.591 |

**Defining "stalled".**

- **Choice of X.** Among buildings with both dates (construction start on or before the snapshot), announcement-to-construction-start has a median of 10.1 months, a 75th percentile of 20.3 and an 80th of 23.0. **X = 24 months** is therefore the point by which about 80% of projects that do break ground have done so.
- **Definition.** `stalled_X` = not failed, campus stage in {Announcement, Delayed, Land Bank}, no construction start by 2026-06-30, and announced more than X months before the snapshot.
- **Coverage.** It is defined only for campuses old enough to be eligible. Delayed is included as stalled. Robustness thresholds are X = 12, 18 and 36.

| outcome | n_defined | n_eq_1 | share | n_defined_with_ann_date | n_eq_1_with_ann_date |
|---|---|---|---|---|---|
| failed (all buildings withdrawn/cancelled) | 2425 | 108 | 0.045 | 1068 | 87 |
| failed_any (any building) | 2425 | 109 | 0.045 | 1068 | 87 |
| stalled_12 | 628 | 256 | 0.408 | 628 | 256 |
| stalled_18 | 428 | 138 | 0.322 | 428 | 138 |
| stalled_24 | 300 | 72 | 0.240 | 300 | 72 |
| stalled_36 | 145 | 20 | 0.138 | 145 | 20 |

**Power on stalled is weak.** Only 300 campuses are old enough for X = 24, and 145 for X = 36.

# 4. Descriptives

## B3. Balance table (the confounding problem)

All 3,143 counties by 2025 QCEW newspaper-establishment group:

| variable | mean_0 | mean_1 | mean_2+ | mean_all | n_nonmissing |
|---|---|---|---|---|---|
| Population (ACS 2017-21) | 23,486.962 | 35,347.432 | 250,731.060 | 104,907.885 | 3143 |
| Density (per sq mi) | 82.939 | 95.428 | 640.303 | 275.948 | 3143 |
| Median HH income ($) | 54,830.187 | 55,050.807 | 64,660.960 | 58,235.708 | 3142 |

| variable | mean_0 | mean_1 | mean_2+ | mean_all | n_nonmissing |
|---|---|---|---|---|---|
| RUCC 2023 (1-9) | 6.210 | 5.850 | 3.792 | 5.249 | 3143 |
| Metro (share) | 0.277 | 0.275 | 0.577 | 0.377 | 3143 |
| Share BA+ | 0.198 | 0.205 | 0.285 | 0.230 | 3143 |
| Share 65+ | 0.199 | 0.197 | 0.181 | 0.192 | 3143 |
| Substations >=230kV | 1.295 | 1.331 | 3.784 | 2.149 | 3143 |
| km of >=345kV lines | 37.540 | 39.448 | 73.235 | 50.348 | 3143 |
| Ag land value $/acre (2022) | 4,216.536 | 6,673.387 | 10,130.158 | 7,219.393 | 3071 |
| Industrial price (c/kWh) | 9.166 | 9.203 | 10.294 | 9.565 | 3090 |
| Baseline water stress (raw) | 0.343 | 0.301 | 0.295 | 0.309 | 3134 |
| Rep. share 2020 | 0.690 | 0.686 | 0.575 | 0.650 | 3112 |
| Rep. share 2024 | 0.710 | 0.705 | 0.595 | 0.669 | 3112 |
| General-purpose govts | 7.744 | 10.524 | 18.003 | 12.325 | 3143 |
| Gen.-purpose govts per 10k | 10.062 | 9.034 | 4.117 | 7.642 | 3143 |
| Full-power TV stations | 0.078 | 0.220 | 1.285 | 0.542 | 3143 |
| Full-power radio stations | 2.061 | 3.232 | 8.300 | 4.637 | 3143 |
| Counties (N) | 816.000 | 1,267.000 | 1,060.000 | 3,143.000 | 3143 |

Counties hosting at least one campus announced 2022+:

| variable | mean_0 | mean_1 | mean_2+ | mean_all | n_nonmissing |
|---|---|---|---|---|---|
| Population (ACS 2017-21) | 55,694.636 | 72,248.480 | 526,422.318 | 326,618.818 | 461 |
| Density (per sq mi) | 211.548 | 151.743 | 899.985 | 585.357 | 461 |
| Median HH income ($) | 66,726.169 | 60,989.374 | 70,275.291 | 67,204.900 | 461 |
| RUCC 2023 (1-9) | 3.870 | 3.959 | 2.165 | 2.928 | 461 |
| Metro (share) | 0.610 | 0.528 | 0.858 | 0.729 | 461 |
| Share BA+ | 0.241 | 0.222 | 0.328 | 0.285 | 461 |
| Share 65+ | 0.178 | 0.173 | 0.158 | 0.165 | 461 |
| Substations >=230kV | 3.701 | 3.000 | 7.352 | 5.581 | 461 |
| km of >=345kV lines | 80.097 | 92.737 | 140.325 | 117.568 | 461 |
| Ag land value $/acre (2022) | 6,015.757 | 5,260.049 | 13,024.019 | 9,788.223 | 453 |
| Industrial price (c/kWh) | 9.301 | 9.719 | 10.268 | 9.965 | 457 |
| Baseline water stress (raw) | 0.406 | 0.322 | 0.344 | 0.349 | 461 |
| Rep. share 2020 | 0.625 | 0.664 | 0.512 | 0.571 | 457 |
| Rep. share 2024 | 0.642 | 0.683 | 0.532 | 0.590 | 457 |
| General-purpose govts | 7.377 | 11.382 | 22.828 | 17.193 | 461 |
| Gen.-purpose govts per 10k | 4.121 | 3.138 | 1.383 | 2.308 | 461 |
| Full-power TV stations | 0.078 | 0.447 | 2.525 | 1.562 | 461 |
| Full-power radio stations | 2.195 | 4.317 | 12.391 | 8.534 | 461 |
| Counties (N) | 77.000 | 123.000 | 261.000 | 461.000 | 461 |

- **News supply is a proxy for county size and rurality.** Zero-newspaper counties are about one-tenth the population of 2+ counties, much more rural, poorer, more Republican and cheaper (land $4,217 vs $10,130 per acre). They have more governments per capita and almost no broadcast stations.
- **Within host counties.** The gap persists, but zero-newspaper hosts are much less rural than zero-newspaper counties generally (61% metro). Data centers that land in “news deserts” land in exurban deserts.
- **Industrial power.** It is cheaper in zero counties (9.2 vs 10.3 c/kWh), as is land.
- **Implication.** Any cross-sectional association between news supply and siting is confounded by exactly the fundamentals that drive siting.

## B1–B2. Projects

| ann_year | Active | Announcement | Construction | Delayed | Failed | Land Bank | Total |
|---|---|---|---|---|---|---|---|
| 2010 | 1 | 0 | 0 | 0 | 0 | 0 | 1 |
| 2011 | 3 | 0 | 0 | 0 | 0 | 0 | 3 |
| 2013 | 1 | 0 | 0 | 0 | 0 | 0 | 1 |
| 2015 | 2 | 0 | 0 | 0 | 0 | 0 | 2 |
| 2016 | 5 | 0 | 0 | 0 | 0 | 0 | 5 |
| 2017 | 8 | 0 | 0 | 0 | 0 | 0 | 8 |
| 2018 | 6 | 0 | 1 | 0 | 0 | 0 | 7 |
| 2019 | 6 | 2 | 0 | 0 | 0 | 0 | 8 |
| 2020 | 9 | 0 | 0 | 1 | 0 | 0 | 10 |
| 2021 | 18 | 2 | 2 | 0 | 4 | 0 | 26 |
| 2022 | 22 | 10 | 10 | 1 | 1 | 0 | 44 |
| 2023 | 27 | 23 | 29 | 2 | 3 | 1 | 85 |
| 2024 | 32 | 100 | 75 | 1 | 16 | 6 | 230 |
| 2025 | 16 | 261 | 70 | 6 | 51 | 5 | 409 |
| 2026 | 4 | 188 | 12 | 0 | 12 | 11 | 227 |
| Total | 160 | 586 | 199 | 11 | 87 | 23 | 1066 |

| mw_bin | Active | Announcement | Construction | Delayed | Failed | Land Bank | Total |
|---|---|---|---|---|---|---|---|
| 1-10 | 3 | 15 | 3 | 0 | 0 | 0 | 21 |
| 10-50 | 19 | 73 | 14 | 0 | 4 | 12 | 122 |
| 50+ | 133 | 376 | 176 | 11 | 58 | 3 | 757 |
| <=1 | 2 | 1 | 0 | 0 | 0 | 0 | 3 |
| Total | 157 | 465 | 193 | 11 | 62 | 15 | 903 |

By news supply in the announcement year (QCEW):

| q_grp | campuses | share_failed | share_active | share_construction | share_announcement | share_ai | share_btm | share_mw50 | share_undisclosed | median_mw | stalled_24 | n_stalled_eligible |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 153 | 0.105 | 0.059 | 0.183 | 0.654 | 0.294 | 0.163 | 0.932 | 0.288 | 351.095 | 0.306 | 36 |
| 1 | 206 | 0.063 | 0.102 | 0.228 | 0.607 | 0.340 | 0.180 | 0.901 | 0.257 | 303.960 | 0.190 | 42 |
| 2+ | 705 | 0.082 | 0.184 | 0.176 | 0.557 | 0.218 | 0.085 | 0.801 | 0.241 | 195.925 | 0.241 | 220 |

| q_pcq | campuses | share_failed | share_active | share_construction | share_announcement | share_ai | share_btm | share_mw50 | share_undisclosed | median_mw | stalled_24 | n_stalled_eligible |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 212 | 0.080 | 0.212 | 0.189 | 0.519 | 0.259 | 0.132 | 0.887 | 0.217 | 296.190 | 0.200 | 90 |
| 2 | 639 | 0.088 | 0.152 | 0.177 | 0.584 | 0.241 | 0.088 | 0.813 | 0.249 | 200.000 | 0.251 | 175 |
| 3 | 161 | 0.068 | 0.075 | 0.224 | 0.634 | 0.273 | 0.137 | 0.852 | 0.342 | 289.400 | 0.308 | 26 |
| 4 | 52 | 0.058 | 0.115 | 0.192 | 0.635 | 0.308 | 0.308 | 0.913 | 0.135 | 398.000 | 0.286 | 7 |

- **Projects in zero-newspaper counties are bigger and more AI/BTM-heavy.**
  - Median 351 MW vs 196 MW in 2+ counties.
  - 93% are above 50 MW.
  - 29% have the AI flag (vs 22%) and 16% the BTM flag (vs 9%).
  - They are more often undisclosed: 29% vs 24%.
- **They fail somewhat more often:** 10.5% vs 8.2% of dated campuses.
- **Per-capita quartiles.** These are dominated by county size. The large metro hosts (Loudoun, Maricopa, Dallas) have few newspapers *per capita* and fall in quartile 2, so the quartile measure does not order counties by news supply in the way the 0/1/2+ grouping does.

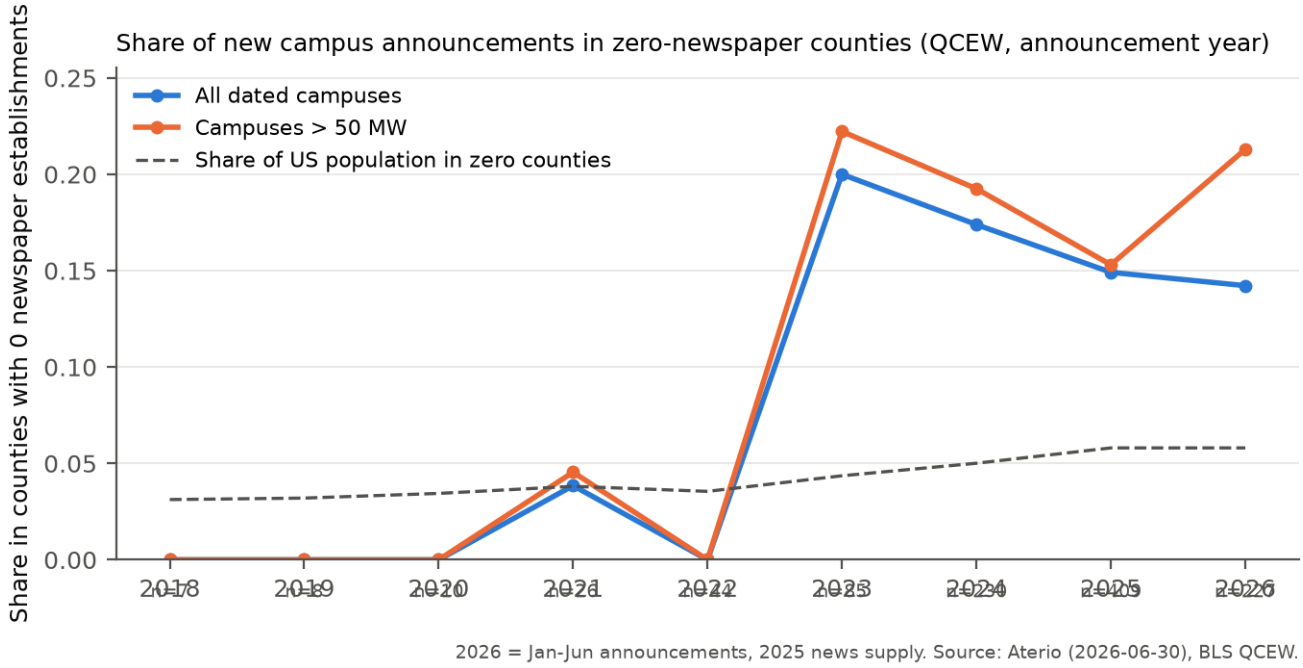



Figure 1: Share of new announcements in zero-newspaper counties

| ann_year | n | share_zero | n_50 | share_zero_50 | share_counties_zero | share_pop_in_zero |
|---|---|---|---|---|---|---|
| 2010.000 | 1.000 | 0.000 | 1.000 | 0.000 | 0.122 | 0.018 |
| 2011.000 | 3.000 | 0.000 | 2.000 | 0.000 | 0.127 | 0.019 |
| 2013.000 | 1.000 | 0.000 | 1.000 | 0.000 | 0.138 | 0.022 |
| 2015.000 | 2.000 | 0.500 | 2.000 | 0.500 | 0.148 | 0.024 |
| 2016.000 | 5.000 | 0.200 | 3.000 | 0.333 | 0.154 | 0.026 |
| 2017.000 | 8.000 | 0.000 | 7.000 | 0.000 | 0.161 | 0.030 |
| 2018.000 | 7.000 | 0.000 | 6.000 | 0.000 | 0.169 | 0.031 |
| 2019.000 | 8.000 | 0.000 | 8.000 | 0.000 | 0.173 | 0.032 |
| 2020.000 | 10.000 | 0.000 | 6.000 | 0.000 | 0.185 | 0.034 |
| 2021.000 | 26.000 | 0.038 | 22.000 | 0.045 | 0.195 | 0.038 |
| 2022.000 | 44.000 | 0.000 | 37.000 | 0.000 | 0.200 | 0.036 |
| 2023.000 | 85.000 | 0.200 | 72.000 | 0.222 | 0.218 | 0.044 |
| 2024.000 | 230.000 | 0.174 | 187.000 | 0.193 | 0.240 | 0.050 |
| 2025.000 | 409.000 | 0.149 | 294.000 | 0.153 | 0.260 | 0.058 |
| 2026.000 | 227.000 | 0.142 | 109.000 | 0.213 | 0.260 | 0.058 |

- **The share of announcements landing in zero-newspaper counties rose** from near 0 in 2017–2022 (n ≤ 44 per year) to 14–20% in 2023–26.
- **It still sits well below the share of counties that are zero (26% in 2025).**
- **It sits well above the share of population living in them (5.8%).**
- **Counts before 2021 are tiny** (n ≤ 10 per year).

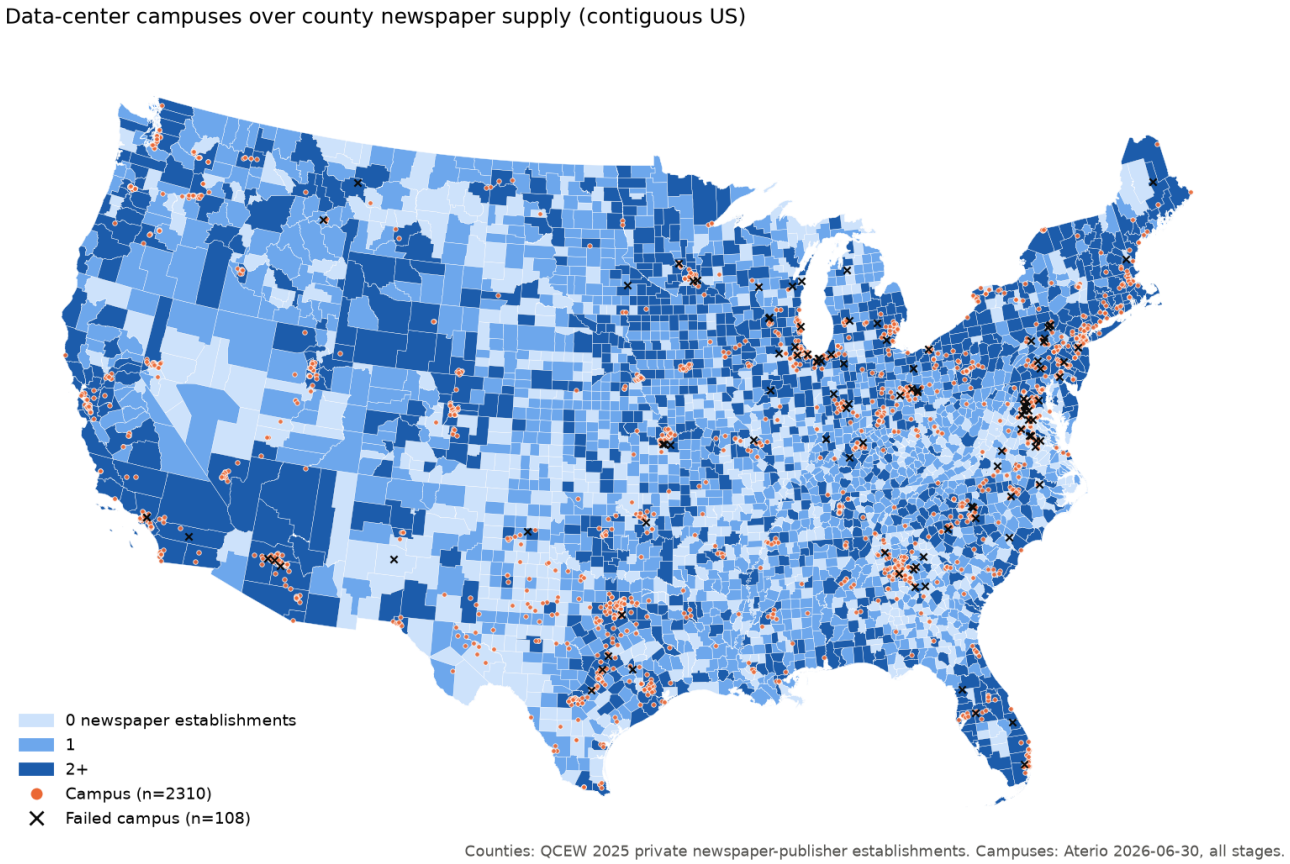



Figure 2: National map

## 5. Regression results

**Notation.** $N_{c,t-1}$ is news supply in county $c$, lagged one year: either `q_pc10k` (QCEW newspaper establishments per 10,000 residents) or `q_zero` (no establishment).

- **County panel (C).**

$$E[A_{ct} \mid \cdot] = \exp\left(\beta N_{c,t-1} + X_c'\gamma + \alpha_c + \delta_{s(c)t}\right)$$

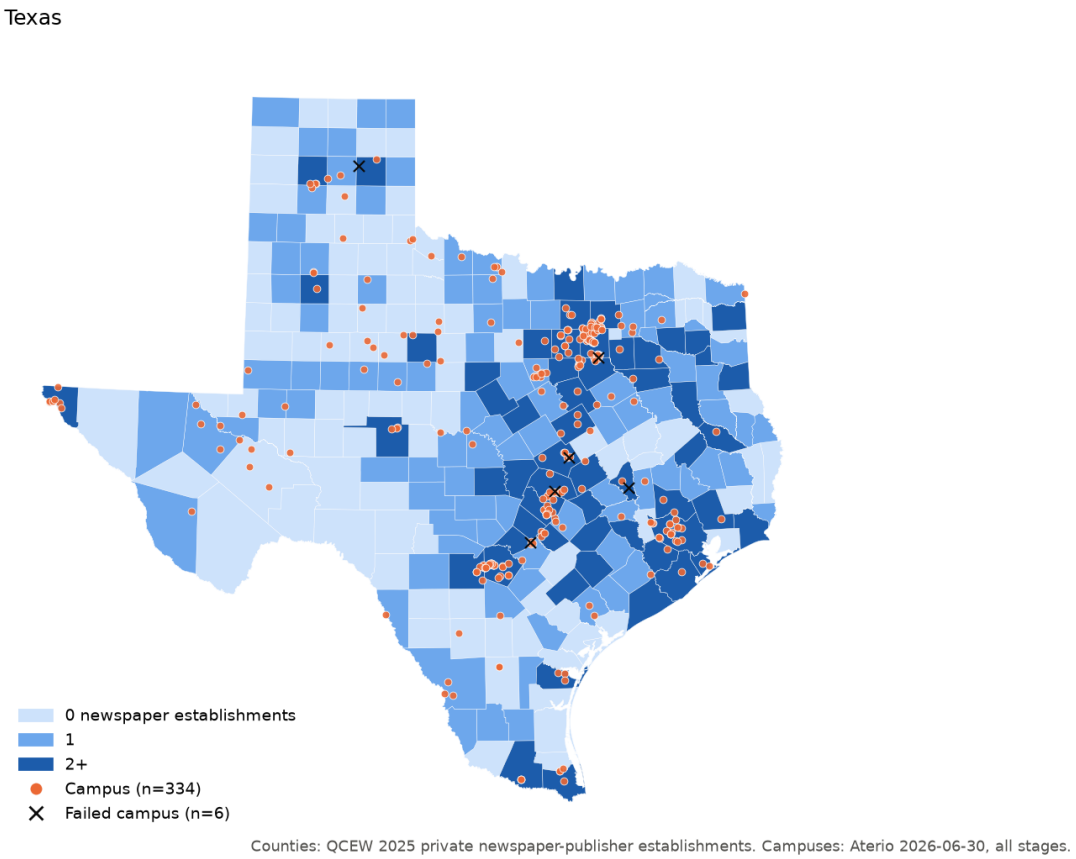



Figure 3: Texas

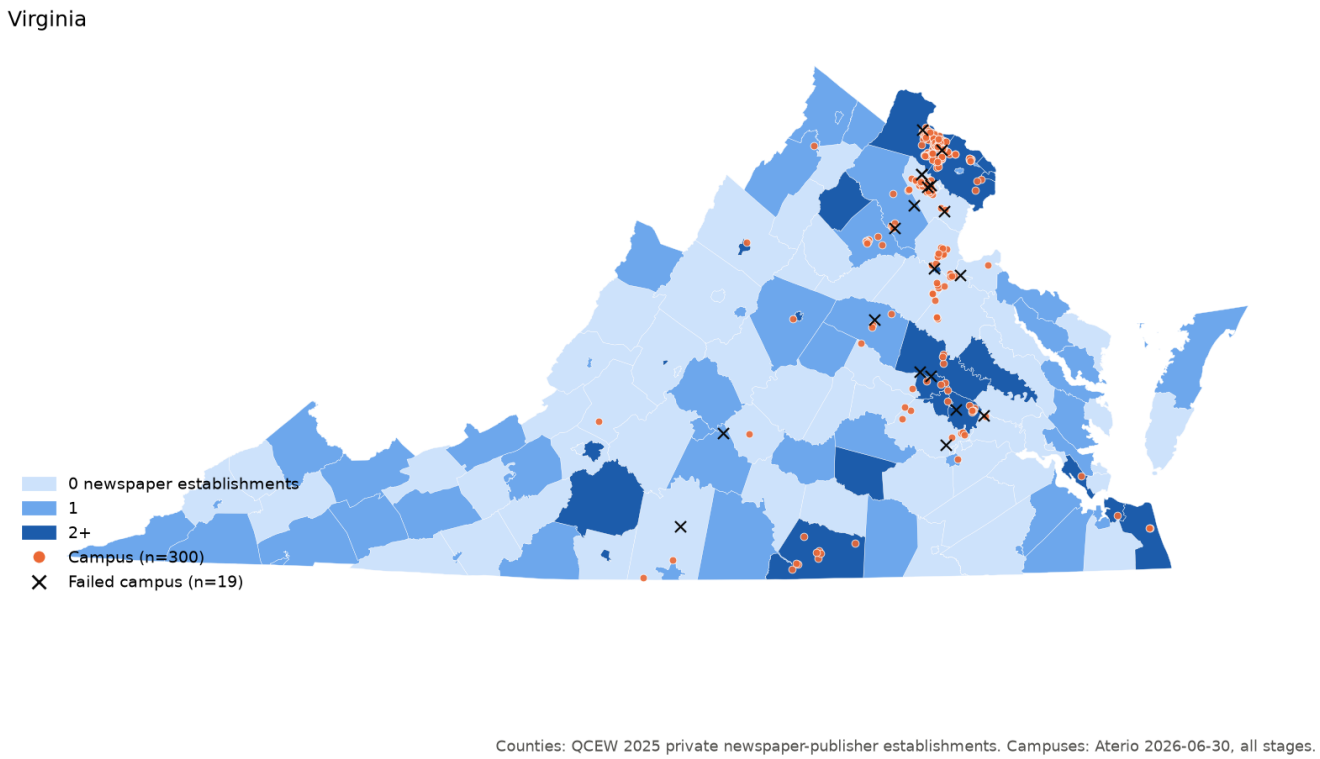



Figure 4: Virginia

estimated by PPML for counts, and as an LPM for $1[A_{ct} > 0]$. SEs are **clustered by state**.

- **Campus level (D, E).**

$$Y_i = \beta N_{c(i),t(i)} + Z_i'\theta + X_{c(i)}'\gamma + \delta_{s\times t} + \varepsilon_i$$

estimated as an LPM, with SEs **clustered by county**.

- **Columns.**
  1. bivariate
  2. demographics + RUCC (+ year FE in C)
  3. siting fundamentals (grid, land value, industrial price, water stress)
  4. state×year FE
  5. C: county FE + state×year FE; D/E: balancing-authority×year FE
- **Coefficient units.** `q_pc10k` coefficients are per one establishment per 10k residents; its 2025 interquartile range across counties is roughly 0.24–0.65.

## C. Extensive margin (county × year, 2010–2026)

- **Panel.** It has 53,431 county-years. 2,674 of 3,143 counties (85%) never have an announcement, and only 716 county-years have one [`43_extensive.txt`].
- **Common sample.** Excluding counties with missing controls leaves 51,663 county-years in 3,039 counties.
- **PPML drops.** PPML drops all-zero groups: N falls to 15,714 with state×year FE and to 2,953 with county FE (the 458 announcing counties' county-years in announcing state-years).

**Announcements (count, PPML):**

| term | (1) bivariate | (2) +demog+RUCC, year FE | (3) +fundamentals | (4) state x year FE | (5) county FE + state x year FE |
|---|---|---|---|---|---|
| q_pc10k_l1 | -2.1551*** (0.3628) | -0.4399* (0.2445) | -0.2130 (0.1776) | -0.1085 (0.1487) | -0.0674 (0.1634) |
| N | 51663 | 45585 | 45585 | 15714 | 2953 |

| term | (1) bivariate | (2) +demog+RUCC, year FE | (3) +fundamentals | (4) state x year FE | (5) county FE + state x year FE |
|---|---|---|---|---|---|
| q_zero_l1 | -0.3010 (0.2389) | -0.0376 (0.1852) | 0.1352 (0.1563) | -0.0630 (0.1613) | -0.1960 (0.2526) |
| N | 51663 | 45585 | 45585 | 15714 | 2953 |

**Any announcement (LPM):**

| term | (1) bivariate | (2) +demog+RUCC, year FE | (3) +fundamentals | (4) state x year FE | (5) county FE + state x year FE |
|---|---|---|---|---|---|
| q_pc10k_l1 | -0.0048*** (0.0010) | 0.0006 (0.0004) | 0.0012** (0.0005) | 0.0015*** (0.0005) | 0.0030*** (0.0011) |
| N | 51663 | 51663 | 51663 | 51663 | 51663 |

| term | (1) bivariate | (2) +demog+RUCC, year FE | (3) +fundamentals | (4) state x year FE | (5) county FE + state x year FE |
|---|---|---|---|---|---|
| q_zero_l1 | -0.0041* (0.0021) | -0.0029 (0.0021) | -0.0021 (0.0023) | -0.0039* (0.0021) | -0.0091** (0.0043) |
| N | 51663 | 51663 | 51663 | 51663 | 51663 |

**MW announced (PPML):**

| term | (1) bivariate | (2) +demog+RUCC, year FE | (3) +fundamentals | (4) state x year FE | (5) county FE + state x year FE |
|---|---|---|---|---|---|
| q_pc10k_l1 | -1.0311** (0.4735) | -0.2089 (0.2472) | -0.0809 (0.2238) | 0.1212 (0.1453) | 0.3215** (0.1575) |
| N | 51663 | 45585 | 45585 | 15157 | 2541 |

| term | (1) bivariate | (2) +demog+RUCC, year FE | (3) +fundamentals | (4) state x year FE | (5) county FE + state x year FE |
|---|---|---|---|---|---|
| q_zero_l1 | 0.0150 (0.2394) | -0.1394 (0.3040) | 0.0328 (0.2992) | -0.3506 (0.3030) | -0.6142* (0.3239) |
| N | 51663 | 45585 | 45585 | 15157 | 2541 |

**What the pattern shows.**

- **Bivariate.** Counties with more newspapers per capita get *fewer* announcements (PPML −2.16). That is the size/rurality composition: per-capita news is high in small rural counties.
- **With controls.** Demographics and RUCC cut the PPML coefficient by 80%, and with state×year FE it is −0.11 (se 0.15).
- **The LPM flips sign.** With state×year FE, and more so with county FE (C5: +0.0030, se 0.0011), more news per capita goes with a higher probability of any announcement. Becoming a zero-newspaper county goes with a 0.9 pp *lower* probability (C5: −0.0091, se 0.0043).
- **Against the baseline.** Mean any-announcement probability is 716/53,431 ≈ 1.3%, so these are not small relative to the mean.
- **The other outcomes in C5.** PPML counts are null. MW announced is positive and significant at 5% for `q_pc10k` (+0.32, se 0.16) and negative, significant only at 10%, for `q_zero` (−0.61, se 0.32).

**What it does not show.**

- **It does not support the extensive-margin sorting hypothesis on these data.** The within-county associations run the opposite way.
- **Where C5 identification comes from.** It rests on 66 announcing counties whose zero status switches and on small per-capita changes (median within-county SD 0.048 among announcing counties) [`c5_within_county_variation`].
- **Per-capita changes are partly mechanical.** A growing county's per-capita count falls even with a constant number of papers. Population growth attracts data centers, which would push the coefficient negative; the C5 sign is positive despite this.
- **The QCEW 2022 code change** also moves within-county counts and is only partly absorbed by state×year FE.

**Heterogeneity** (PPML, stacked type panels; interaction = difference for the type):

| het | regressor | spec | term | coef | se | p | n | note |
|---|---|---|---|---|---|---|---|---|
| mw50 | q_pc10k_l1 | (4) state x year x type FE | q_pc10k_l1 | -0.896 | 0.286 | 0.002 | 20983 | |
| mw50 | q_pc10k_l1 | (4) state x year x type FE | q_pc10k_l1:typ | 0.850 | 0.271 | 0.002 | 20983 | |
| mw50 | q_pc10k_l1 | (5) county x type + state x year x type FE | q_pc10k_l1 | -4.495 | 7.403 | 0.544 | 2526 | |
| mw50 | q_pc10k_l1 | (5) county x type + state x year x type FE | q_pc10k_l1:typ | 4.559 | 7.385 | 0.537 | 2526 | |
| mw50 | q_zero_l1 | (4) state x year x type FE | q_zero_l1 | -0.624 | 0.396 | 0.115 | 20983 | |
| mw50 | q_zero_l1 | (4) state x year x type FE | q_zero_l1:typ | 0.678 | 0.370 | 0.067 | 20983 | |
| mw50 | q_zero_l1 | (5) county x type + state x year x type FE | q_zero_l1 | 12.842 | nan | nan | 2526 | SUSPECT: likely separation / non-convergence |
| mw50 | q_zero_l1 | (5) county x type + state x year x type FE | q_zero_l1:typ | -12.917 | 0.265 | 0.000 | 2526 | SUSPECT: likely separation / non-convergence |
| ai | q_pc10k_l1 | (4) state x year x type FE | q_pc10k_l1 | -0.200 | 0.141 | 0.157 | 22894 | |
| ai | q_pc10k_l1 | (4) state x year x type FE | q_pc10k_l1:typ | 0.263 | 0.261 | 0.313 | 22894 | |
| ai | q_pc10k_l1 | (5) county x type + state x year x type FE | q_pc10k_l1 | -0.222 | 0.240 | 0.355 | 2909 | |
| ai | q_pc10k_l1 | (5) county x type + state x year x type FE | q_pc10k_l1:typ | 0.306 | 0.305 | 0.315 | 2909 | |
| ai | q_zero_l1 | (4) state x year x type FE | q_zero_l1 | -0.179 | 0.167 | 0.284 | 22894 | |
| ai | q_zero_l1 | (4) state x year x type FE | q_zero_l1:typ | 0.506 | 0.222 | 0.023 | 22894 | |
| ai | q_zero_l1 | (5) county x type + state x year x type FE | q_zero_l1 | -0.029 | 0.298 | 0.923 | 2909 | |
| ai | q_zero_l1 | (5) county x type + state x year x type FE | q_zero_l1:typ | -1.557 | 1.154 | 0.177 | 2909 | |
| btm | q_pc10k_l1 | (4) state x year x type FE | q_pc10k_l1 | -0.306 | 0.157 | 0.052 | 19863 | |
| btm | q_pc10k_l1 | (4) state x year x type FE | q_pc10k_l1:typ | 0.688 | 0.140 | 0.000 | 19863 | |
| btm | q_pc10k_l1 | (5) county x type + state x year x type FE | q_pc10k_l1 | -0.136 | 0.168 | 0.420 | 2843 | |
| btm | q_pc10k_l1 | (5) county x type + state x year x type FE | q_pc10k_l1:typ | 0.402 | 0.172 | 0.019 | 2843 | |
| btm | q_zero_l1 | (4) state x year x type FE | q_zero_l1 | -0.129 | 0.190 | 0.498 | 19863 | |
| btm | q_zero_l1 | (4) state x year x type FE | q_zero_l1:typ | 0.538 | 0.365 | 0.140 | 19863 | |
| btm | q_zero_l1 | (5) county x type + state x year x type FE | q_zero_l1 | -0.246 | 0.260 | 0.344 | 2843 | |
| btm | q_zero_l1 | (5) county x type + state x year x type FE | q_zero_l1:typ | 0.215 | 0.508 | 0.672 | 2843 | |
| post2021 | q_pc10k_l1 | (4) state x year FE | q_pc10k_l1 | -1.384 | 0.705 | 0.050 | 15714 | |
| post2021 | q_pc10k_l1 | (4) state x year FE | q_pc10k_l1:post2021 | 1.323 | 0.764 | 0.083 | 15714 | |
| post2021 | q_pc10k_l1 | (5) county FE + state x year FE | q_pc10k_l1 | -1.471 | 0.838 | 0.079 | 2953 | |
| post2021 | q_pc10k_l1 | (5) county FE + state x year FE | q_pc10k_l1:post2021 | 1.604 | 0.938 | 0.087 | 2953 | |
| post2021 | q_zero_l1 | (4) state x year FE | q_zero_l1 | -1.254 | 0.810 | 0.121 | 15714 | |
| post2021 | q_zero_l1 | (4) state x year FE | q_zero_l1:post2021 | 1.229 | 0.774 | 0.112 | 15714 | |
| post2021 | q_zero_l1 | (5) county FE + state x year FE | q_zero_l1 | -1.447 | 0.915 | 0.114 | 2953 | |
| post2021 | q_zero_l1 | (5) county FE + state x year FE | q_zero_l1:post2021 | 1.294 | 0.846 | 0.126 | 2953 | |

- **Size, AI and BTM.** With state×year FE, the negative association of news supply with announcements is

concentrated in projects ≤50 MW. The interaction for >50 MW (+0.85) offsets it, and BTM projects show a similar positive interaction (+0.69; also +0.40 with county FE).
- **Post-2021 (P4).** The coefficient differs (`q_pc10k`: −1.38 pre, +1.32 interaction with state×year FE), but this is identified off very few pre-2022 announcements, and the interaction is insignificant at 5%.
- **One cell failed to converge.** The q_zero × 50+ MW cell with county FE (PPML separation) is marked `[suspect]` / `SUSPECT` and should be ignored.
- **The predicted sign is absent.** None of this shows sorting into low-news places becoming *stronger* after the AI boom.

**Event study (C6).**

- **Design.** The stack is built around a county's first decline in QCEW newspaper establishments in cohort years 2008–2023. Controls are counties in the same state and same lagged size band with no decline in the window. Outcome = any announcement; reference period k = −1.

| cohort | treated_counties | matched_treated | controls | trt_cy_ann | trt_cty_ann |
|---|---|---|---|---|---|
| 2008 | 200 | 200 | 1058 | 1 | 1 |
| 2009 | 194 | 194 | 1127 | 0 | 0 |
| 2010 | 160 | 160 | 883 | 0 | 0 |
| 2011 | 155 | 155 | 892 | 0 | 0 |
| 2012 | 154 | 154 | 1094 | 0 | 0 |
| 2013 | 151 | 151 | 1027 | 0 | 0 |
| 2014 | 149 | 149 | 948 | 2 | 2 |
| 2015 | 147 | 147 | 1159 | 1 | 1 |
| 2016 | 155 | 155 | 942 | 4 | 4 |
| 2017 | 184 | 184 | 1105 | 0 | 0 |
| 2018 | 209 | 209 | 1201 | 7 | 6 |
| 2019 | 182 | 182 | 1228 | 3 | 2 |
| 2020 | 183 | 183 | 1097 | 7 | 7 |
| 2021 | 167 | 167 | 1145 | 20 | 13 |
| 2022 | 160 | 160 | 1129 | 26 | 24 |
| 2023 | 227 | 227 | 1310 | 74 | 41 |

| term | coef | se | p | n | k | trt_cy_ann |
|---|---|---|---|---|---|---|
| ev_0 | -0.001 | 0.001 | 0.188 | 122809 | -3 | 145 |
| ev_1 | 0.000 | 0.001 | 0.709 | 122809 | -2 | 145 |
| ref | 0.000 | 0.000 | nan | 122809 | -1 | 145 |
| ev_3 | 0.002 | 0.001 | 0.097 | 122809 | 0 | 145 |
| ev_4 | 0.005 | 0.002 | 0.010 | 122809 | 1 | 145 |
| ev_5 | 0.007 | 0.002 | 0.002 | 122809 | 2 | 145 |
| ev_6 | 0.011 | 0.003 | 0.001 | 122809 | 3 | 145 |

- **Column key.** `trt_cy_ann` = treated county-years with an announcement; `trt_cty_ann` = treated counties with any announcement in the window.
- **Data.** There are 145 treated county-years with an announcement, almost all in the 2021–23 cohorts.
- **Pre-trends.** They are flat: k = −3: −0.0015 (se 0.0011); k = −2: +0.0004.
- **Post-event.** Coefficients rise steadily to +0.011 at k = +3 (se 0.003).
- **Interpretation.**
  - **Why not as a deterrence or sorting effect.** Treated counties are those large enough to lose an establishment, and the post periods of the 2021–23 cohorts are the AI-boom years. Size-band matching (0–1 / 2 / 3–5 / 6+) does not equalize Loudoun with a 6-paper rural county.
  - **What it shows.** Declines happen where data-center growth happens. Nothing here isolates the news channel.
- **1→0 version.** It was not estimated: only 23 treated county-years have an announcement [`c6_event_cohorts_1to0`].

## D. Intensive margin (campus level)

- **Sample.** It has 1,049 dated campuses with controls. 87 failed, and 299 are eligible for `stalled_24` (72 stalled).
- **Controls.** Project controls are log MW (with missing flag), AI, BTM, and provider public. County and siting controls are demographics, RUCC, county grid (substations ≥230 kV, km ≥345 kV lines), land value, county industrial price, water stress, project distance to a ≥230 kV substation, project water stress and the utility industrial price (missing-indicator method).
- **SEs** are clustered by county.

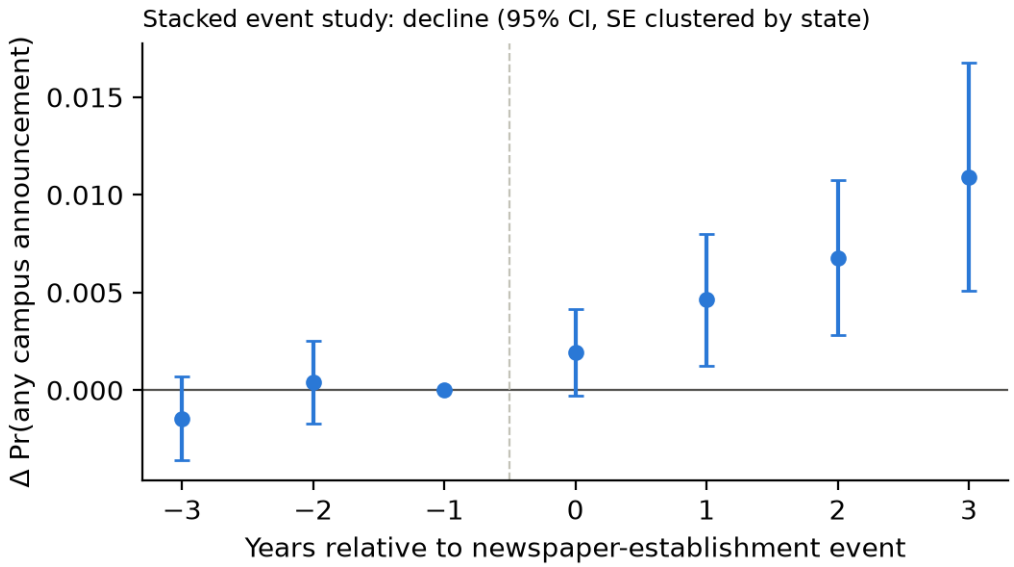



Figure 5: Event study, declines

**Failed:**

| term | (1) bivariate | (2) +project | (3) +county & siting controls | (4) state x year FE | (5) BA x year FE |
|---|---|---|---|---|---|
| q_pc10k | -0.0261 (0.0264) | -0.0216 (0.0285) | -0.0250 (0.0317) | -0.0347 (0.0273) | -0.0214 (0.0374) |
| N | 1049 | 1049 | 1049 | 968 | 719 |

| term | (1) bivariate | (2) +project | (3) +county & siting controls | (4) state x year FE | (5) BA x year FE |
|---|---|---|---|---|---|
| q_zero | 0.0296 (0.0288) | 0.0280 (0.0281) | 0.0238 (0.0305) | 0.0288 (0.0360) | 0.0189 (0.0400) |
| N | 1049 | 1049 | 1049 | 968 | 719 |

**Stalled (X = 24):**

| term | (1) bivariate | (2) +project | (3) +county & siting controls | (4) state x year FE | (5) BA x year FE |
|---|---|---|---|---|---|
| q_pc10k | 0.0454 (0.1721) | 0.0651 (0.1756) | 0.0719 (0.1623) | 0.3709 (0.3279) | 0.2544 (0.2348) |
| N | 298 | 298 | 298 | 232 | 228 |

| term | (1) bivariate | (2) +project | (3) +county & siting controls | (4) state x year FE | (5) BA x year FE |
|---|---|---|---|---|---|
| q_zero | 0.0727 (0.0751) | 0.1026 (0.0758) | 0.0921 (0.0713) | -0.0019 (0.1163) | 0.0259 (0.0894) |
| N | 298 | 298 | 298 | 232 | 228 |

- **Failure.** Failure is not associated with either news measure in any column.
  - Point estimates are stable: −0.02 to −0.04 for `q_pc10k`, +0.02 to +0.03 for `q_zero`.
  - Precision is poor: with 87 failures, the zero-county coefficient's 95% CI spans roughly −4 to +10 pp, against a failed share of 8.2% in 2+ counties.
  - The signs are the *opposite* of "more news → more failures".
- **Stalled.**
  - **Unstable across thresholds.** `q_pc10k` is insignificant in every column at X = 24 (largest +0.37, se 0.33, with state×year FE; it was +0.51, se 0.25, p = 0.046, before grid controls were added). `q_zero` is positive at X = 12 (about +0.15 to +0.21, significant), insignificant at X = 24 and X = 36, and `q_pc10k` is strongly negative at X = 36 (−1.17, se 0.50, with state×year FE; 88 campuses).
  - **No consistent sign** emerges across thresholds [`d_robustness_outcomes_coef`, `d_all_long`].

| regressor | spec | hazard_ratio | coef | se | p | n | events |
|---|---|---|---|---|---|---|---|
| q_pc10k | (1) bivariate | 1.260 | 0.231 | 0.167 | 0.166 | 1049 | 348 |
| q_pc10k | (2) +project | 1.123 | 0.116 | 0.159 | 0.465 | 1049 | 348 |
| q_pc10k | (3) +county & siting | 1.259 | 0.231 | 0.147 | 0.117 | 1049 | 348 |
| q_zero | (1) bivariate | 0.799 | -0.225 | 0.163 | 0.169 | 1049 | 348 |
| q_zero | (2) +project | 0.679 | -0.387 | 0.165 | 0.019 | 1049 | 348 |
| q_zero | (3) +county & siting | 0.665 | -0.408 | 0.179 | 0.023 | 1049 | 348 |

**Time to construction start (Cox).**

- **Result.** Zero-newspaper hosts are associated with *slower* starts (HR 0.67, p = 0.02 with controls). That matches the stall pattern at X = 12.
- **Caveats.** Construction starts are observed for only 348 campuses, Cox SEs are not clustered, and there are no state×year FE.

**Place level.** This is not possible. QCEW has no place or ZIP detail, and CBP ZIP files censor <3 cells the same way (243 published ZIP×511110 cells nationally in 2023).

**Within-county.**

| metric | value |
|---|---|
| counties with >=2 dated campuses | 175 |
| … announced in >=2 different years | 139 |
| … with any within-county change in q_pc10k | 93 |
| … with q_zero switching | 3 |
| campuses in usable counties | 520 |
| failed campuses in usable counties | 36 |

| term | coef | se | p | n | spec | note | outcome | regressor | formula | cluster |
|---|---|---|---|---|---|---|---|---|---|---|
| q_pc10k | -0.012 | 0.190 | 0.951 | 514 | county FE + year FE | | failed | q_pc10k | failed ~ q_pc10k + log_mw_f + mw_miss + ai + btm + prov_pub + prov_pub_miss | fips + ann_year |
| q_pc10k | -0.186 | 0.165 | 0.263 | 520 | county FE | | failed | q_pc10k | failed ~ q_pc10k + log_mw_f + mw_miss + ai + btm + prov_pub + prov_pub_miss | fips |
| q_zero | -0.007 | 0.045 | 0.882 | 514 | county FE + year FE | | failed | q_zero | failed ~ q_zero + log_mw_f + mw_miss + ai + btm + prov_pub + prov_pub_miss | fips + ann_year |
| q_zero | -0.008 | 0.056 | 0.881 | 520 | county FE | | failed | q_zero | failed ~ q_zero + log_mw_f + mw_miss + ai + btm + prov_pub + prov_pub_miss | fips |
| q_pc10k | -0.093 | 0.772 | 0.905 | 193 | county FE + year FE | | stalled_24 | q_pc10k | stalled_24 ~ q_pc10k + log_mw_f + mw_miss + ai + btm + prov_pub + prov_pub_miss | fips + ann_year |
| q_pc10k | -0.162 | 0.718 | 0.823 | 198 | county FE | | stalled_24 | q_pc10k | stalled_24 ~ q_pc10k + log_mw_f + mw_miss + ai + btm + prov_pub + prov_pub_miss | fips |
| q_zero | -0.111 | 0.066 | 0.099 | 193 | county FE + year FE | | stalled_24 | q_zero | stalled_24 ~ q_zero + log_mw_f + mw_miss + ai + btm + prov_pub + prov_pub_miss | fips + ann_year |
| q_zero | 0.025 | 0.035 | 0.473 | 198 | county FE | | stalled_24 | q_zero | stalled_24 ~ q_zero + log_mw_f + mw_miss + ai + btm + prov_pub + prov_pub_miss | fips |

- **Usable counties.** There are 139 counties with dated campuses in at least two different years. 93 have any within-county change in per-capita news, which is mostly population-driven, and only 3 switch zero status.
- **Result.** All estimates are null and uninformative.

## E. Secrecy test (P3)

- **Variant check.** "Company Not Disclosed" is the only non-disclosure variant in `PROVIDER_NAME`, after searching for disclos/unknown/confidential/unnamed/TBD/stealth patterns. It covers 1,349 US facilities (18.9%).
- **Sample.** Dated campuses: 25.0% undisclosed and 91.6% with no listed end user.

| term | (1) bivariate | (2) +project | (3) +county controls | (4) state x year FE | (5) BA x year FE |
|---|---|---|---|---|---|
| q_pc10k | -0.0397 (0.0372) | -0.0554 (0.0372) | -0.0877* (0.0468) | -0.0566 (0.0426) | -0.0652 (0.0483) |
| N | 1059 | 1059 | 1059 | 979 | 724 |

| term | (1) bivariate | (2) +project | (3) +county controls | (4) state x year FE | (5) BA x year FE |
|---|---|---|---|---|---|
| q_zero | 0.0502 (0.0488) | 0.0450 (0.0479) | 0.0653 (0.0509) | 0.0505 (0.0563) | 0.0785 (0.0572) |
| N | 1059 | 1059 | 1059 | 979 | 724 |

| term | (1) bivariate | (2) +project | (3) +county controls | (4) state x year FE | (5) BA x year FE |
|---|---|---|---|---|---|

| term | (1) bivariate | (2) +project | (3) +county controls | (4) state x year FE | (5) BA x year FE |
|---|---|---|---|---|---|
| q_pc10k | -0.0676 (0.0412) | -0.0550 (0.0384) | -0.0601 (0.0438) | -0.0844** (0.0386) | -0.0458 (0.0534) |
| N | 1059 | 1059 | 1059 | 979 | 724 |

- **Result.** P3 predicts undisclosed developers are relatively *more* common where news is strong. The estimates point the other way, and weakly.
  - `q_pc10k` is negative in all columns (–0.04 to –0.09), significant only in column 3 and only at 10% (–0.088, se 0.047, p = 0.06).
  - `q_zero` is positive (+0.05 to +0.08), never significant.
- **All-campus sample.** It uses 2025 news and gives the same sign [`e_all_long`].

**Is "undisclosed" developer secrecy or Aterio ignorance?**

| index | Jun CND=0 | Jun CND=1 | Jun CND=All |
|---|---|---|---|
| Feb CND=0 | 4438 | 0 | 4438 |
| Feb CND=1 | 331 | 998 | 1329 |
| Feb CND=All | 4769 | 998 | 5767 |

| q_grp | cnd_in_feb | resolved_by_jun | share_resolved |
|---|---|---|---|
| 0 | 308 | 84 | 0.273 |
| 1 | 380 | 94 | 0.247 |
| 2+ | 641 | 153 | 0.239 |

- **Resolution.** Between Feb and Jun 2026, 331 of 1,329 undisclosed facilities (25%) gained a named provider, and none went the other way. Resolution rates are similar across news groups (27% zero, 25% one, 24% 2+).
- **What resolution looks like.** Disclosure often happens as projects advance, consistent with real developer secrecy (NDAs, code names) that later lifts. That is not proof.
- **Detection lag.** Undisclosed status is not associated with it (coef 0.13 months, se 0.33) [`e_detect_lag_regression`].
- **Undated campuses.** Undisclosed campuses are rarely undated (12% vs 62%), because they are recent announcements.

| state | campuses | undisclosed_n | undisclosed_share |
|---|---|---|---|
| TX | 340 | 32 | 0.094 |
| VA | 319 | 45 | 0.141 |
| CA | 177 | 9 | 0.051 |
| GA | 130 | 43 | 0.331 |
| OH | 111 | 8 | 0.072 |
| IL | 104 | 11 | 0.106 |
| PA | 90 | 35 | 0.389 |
| AZ | 82 | 7 | 0.085 |
| NY | 73 | 12 | 0.164 |
| FL | 66 | 6 | 0.091 |
| NC | 57 | 8 | 0.140 |
| IN | 53 | 8 | 0.151 |
| OR | 52 | 2 | 0.038 |
| WA | 50 | 1 | 0.020 |
| NJ | 48 | 5 | 0.104 |
| MN | 48 | 4 | 0.083 |
| MI | 45 | 3 | 0.067 |
| MO | 44 | 8 | 0.182 |
| CO | 38 | 2 | 0.053 |
| UT | 37 | 3 | 0.081 |
| NV | 36 | 3 | 0.083 |
| WI | 35 | 4 | 0.114 |
| IA | 33 | 1 | 0.030 |
| TN | 32 | 2 | 0.062 |
| OK | 31 | 3 | 0.097 |
| MA | 22 | 0 | 0.000 |
| MD | 21 | 3 | 0.143 |
| SC | 21 | 3 | 0.143 |
| KY | 20 | 5 | 0.250 |

| state | campuses | undisclosed_n | undisclosed_share |
|---|---|---|---|
| KS | 20 | 4 | 0.200 |
| AL | 19 | 0 | 0.000 |
| LA | 16 | 0 | 0.000 |
| CT | 16 | 1 | 0.062 |
| NE | 15 | 1 | 0.067 |
| WY | 14 | 0 | 0.000 |
| MS | 13 | 1 | 0.077 |
| NM | 10 | 2 | 0.200 |
| MT | 10 | 2 | 0.200 |
| ME | 10 | 5 | 0.500 |
| AR | 9 | 2 | 0.222 |
| WV | 9 | 0 | 0.000 |
| DE | 8 | 2 | 0.250 |
| NH | 7 | 1 | 0.143 |
| ID | 7 | 0 | 0.000 |
| ND | 6 | 1 | 0.167 |
| DC | 6 | 0 | 0.000 |
| SD | 5 | 2 | 0.400 |
| AK | 4 | 3 | 0.750 |
| RI | 3 | 0 | 0.000 |
| HI | 2 | 0 | 0.000 |
| VT | 1 | 0 | 0.000 |

**State differences are large.** PA (39%), GA (33%) and NY (16%) have high undisclosed shares; WA (2%), OR (4%) and CA (5%) have low ones. State norms on NDAs and public records plausibly dominate, which column 4 absorbs.

## F. Placebos and leave-one-state-out

**F1. Warehousing (NAICS 493) placebo.** Outcomes are new warehousing establishments: net new (PPML) and any increase (LPM).

| outcome | regressor | spec | coef | se | p | n |
|---|---|---|---|---|---|---|
| wh_net_new | q_pc10k_l1 | (2) +demog+RUCC, year FE | -0.270 | 0.140 | 0.054 | 48624 |
| wh_net_new | q_pc10k_l1 | (3) +fundamentals | -0.111 | 0.069 | 0.107 | 48624 |
| wh_net_new | q_pc10k_l1 | (4) state x year FE | -0.099 | 0.072 | 0.172 | 48035 |
| wh_net_new | q_pc10k_l1 | (5) county FE + state x year FE | -0.007 | 0.090 | 0.942 | 32134 |
| wh_net_new | q_zero_l1 | (2) +demog+RUCC, year FE | -0.610 | 0.120 | 0.000 | 48624 |
| wh_net_new | q_zero_l1 | (3) +fundamentals | -0.405 | 0.083 | 0.000 | 48624 |
| wh_net_new | q_zero_l1 | (4) state x year FE | -0.312 | 0.082 | 0.000 | 48035 |
| wh_net_new | q_zero_l1 | (5) county FE + state x year FE | -0.011 | 0.103 | 0.917 | 32134 |
| wh_any_new | q_pc10k_l1 | (2) +demog+RUCC, year FE | 0.002 | 0.002 | 0.413 | 48624 |
| wh_any_new | q_pc10k_l1 | (3) +fundamentals | 0.004 | 0.002 | 0.041 | 48624 |
| wh_any_new | q_pc10k_l1 | (4) state x year FE | 0.003 | 0.002 | 0.043 | 48624 |
| wh_any_new | q_pc10k_l1 | (5) county FE + state x year FE | -0.001 | 0.001 | 0.334 | 48624 |
| wh_any_new | q_zero_l1 | (2) +demog+RUCC, year FE | -0.022 | 0.009 | 0.016 | 48624 |
| wh_any_new | q_zero_l1 | (3) +fundamentals | -0.021 | 0.007 | 0.004 | 48624 |
| wh_any_new | q_zero_l1 | (4) state x year FE | -0.010 | 0.006 | 0.121 | 48624 |
| wh_any_new | q_zero_l1 | (5) county FE + state x year FE | 0.001 | 0.007 | 0.904 | 48624 |
| n_ann (same sample 2010-2025) | q_pc10k_l1 | (4) state x year FE | -0.045 | 0.156 | 0.771 | 12906 |
| n_ann (same sample 2010-2025) | q_pc10k_l1 | (5) county FE + state x year FE | -0.092 | 0.118 | 0.433 | 2054 |
| n_ann (same sample 2010-2025) | q_zero_l1 | (4) state x year FE | -0.060 | 0.183 | 0.745 | 12906 |
| n_ann (same sample 2010-2025) | q_zero_l1 | (5) county FE + state x year FE | -0.177 | 0.246 | 0.471 | 2054 |
| any_ann (same sample 2010-2025) | q_pc10k_l1 | (4) state x year FE | 0.001 | 0.000 | 0.004 | 48624 |
| any_ann (same sample 2010-2025) | q_pc10k_l1 | (5) county FE + state x year FE | 0.002 | 0.001 | 0.011 | 48624 |
| any_ann (same sample 2010-2025) | q_zero_l1 | (4) state x year FE | -0.004 | 0.001 | 0.014 | 48624 |
| any_ann (same sample 2010-2025) | q_zero_l1 | (5) county FE + state x year FE | -0.007 | 0.004 | 0.063 | 48624 |

- **Cross-sectional associations are shared with warehousing.** In columns 2–4, zero-newspaper counties get significantly fewer new warehouses (PPML −0.31 to −0.61), mirroring the data-center pattern. That suggests the cross-sectional associations reflect rurality and market size rather than scrutiny.
- **The C5 data-center association is not shared.** With county + state×year FE the warehousing coefficients are 0 (−0.007 and −0.011). The data-center LPM coefficients in the same 2010–2025 sample remain (`q_pc10k` +0.0025, se 0.0009; `q_zero` −0.0070, se 0.0037).

**F2. Placebo timing** (news measured 5 years *after*).

- **Panel version.** It is feasible only for outcome years ≤ 2020, which contain 45 announcements in 42 county-years. All coefficients are about 0, as are the real-timing ones in that window [`f2_placebo_timing`], so this placebo has no power.

- **Campus version.** It was not run: 47 campuses were announced ≤ 2020 and none failed.

**F3. Leave-one-state-out.**

| model | term | dropped | coef | se | p | n |
|---|---|---|---|---|---|---|
| C5 any_ann LPM | q_pc10k_l1 | none | 0.003 | 0.001 | 0.009 | 51663 |
| C5 any_ann LPM | q_pc10k_l1 | VA | 0.003 | 0.001 | 0.010 | 49997 |
| C5 any_ann LPM | q_pc10k_l1 | TX | 0.002 | 0.001 | 0.007 | 47447 |
| C5 any_ann LPM | q_pc10k_l1 | VA+TX | 0.002 | 0.001 | 0.008 | 45781 |
| C5 any_ann LPM | q_zero_l1 | none | -0.009 | 0.004 | 0.042 | 51663 |
| C5 any_ann LPM | q_zero_l1 | VA | -0.010 | 0.004 | 0.024 | 49997 |
| C5 any_ann LPM | q_zero_l1 | TX | -0.005 | 0.003 | 0.051 | 47447 |
| C5 any_ann LPM | q_zero_l1 | VA+TX | -0.006 | 0.003 | 0.019 | 45781 |
| C5 n_ann PPML | q_pc10k_l1 | none | -0.067 | 0.163 | 0.680 | 2953 |
| C5 n_ann PPML | q_pc10k_l1 | VA | -0.097 | 0.176 | 0.582 | 2689 |
| C5 n_ann PPML | q_pc10k_l1 | TX | -0.997 | 0.830 | 0.230 | 2084 |
| C5 n_ann PPML | q_pc10k_l1 | VA+TX | -1.704 | 0.883 | 0.054 | 1820 |
| C5 n_ann PPML | q_zero_l1 | none | -0.196 | 0.253 | 0.438 | 2953 |
| C5 n_ann PPML | q_zero_l1 | VA | 0.023 | 0.458 | 0.960 | 2689 |
| C5 n_ann PPML | q_zero_l1 | TX | -0.148 | 0.306 | 0.630 | 2084 |
| C5 n_ann PPML | q_zero_l1 | VA+TX | 0.227 | 0.630 | 0.718 | 1820 |
| D4 failed LPM | q_pc10k | none | -0.035 | 0.027 | 0.192 | 970 |
| D4 failed LPM | q_pc10k | VA | -0.020 | 0.024 | 0.411 | 809 |
| D4 failed LPM | q_pc10k | TX | -0.125 | 0.075 | 0.098 | 783 |
| D4 failed LPM | q_pc10k | VA+TX | -0.081 | 0.086 | 0.348 | 622 |
| D4 failed LPM | q_zero | none | 0.028 | 0.036 | 0.432 | 970 |
| D4 failed LPM | q_zero | VA | 0.026 | 0.042 | 0.537 | 809 |
| D4 failed LPM | q_zero | TX | 0.034 | 0.042 | 0.419 | 783 |
| D4 failed LPM | q_zero | VA+TX | 0.029 | 0.054 | 0.597 | 622 |

| model | term | min_coef | max_coef | share_p_lt_05 | n_states |
|---|---|---|---|---|---|
| C5 any_ann LPM | q_pc10k_l1 | 0.002 | 0.004 | 1.000 | 49 |
| C5 any_ann LPM | q_zero_l1 | -0.011 | -0.005 | 0.776 | 49 |
| C5 n_ann PPML | q_pc10k_l1 | -0.997 | 0.019 | 0.000 | 49 |
| C5 n_ann PPML | q_zero_l1 | -0.341 | 0.023 | 0.000 | 49 |
| D4 failed LPM | q_pc10k | -0.125 | -0.020 | 0.000 | 46 |
| D4 failed LPM | q_zero | 0.017 | 0.046 | 0.000 | 46 |

- **C5 LPM.** The coefficients keep their sign under every single-state drop. `q_pc10k` is significant at 5% in 49 of 49 drops and `q_zero` in 38 of 49. Dropping VA and TX together leaves them significant (+0.0022, −0.0062).
- **C5 PPML.** It stays null.
- **D4 failure.** It stays null in all drops; dropping TX makes `q_pc10k` −0.12 (se 0.08).

# 6. What these diagnostics cannot tell us

- **Establishment counts ≠ newsroom capacity or coverage of a project.** A county can have one newspaper establishment that never covered the rezoning, or none and be covered by a regional daily or TV station. The relevant treatment, coverage of *this* project before the decision, is not measured anywhere in these data.
- **Aterio outcomes ≠ local decisions.**
  - "Not Approved/Withdrawn" mixes denials, developer withdrawals, utility-capacity failures and financing collapses.
  - There are no hearing dates and no vote records.
  - Stage is a snapshot, so "stalled" conflates slow permitting with shelved projects.
- **Selection into Aterio.**
  - Undetected projects are unobservable.
  - The database began in mid-2024, so pre-2024 failures may be under-recorded.
  - Old active facilities mostly lack announcement dates, so the dated sample is essentially the 2023–26 AI-era cohort.
- **No measure of pre-hearing coverage, and no hearing dates.** The causal story runs through public attention before a decision, and nothing here observes that timing.
- **Thin treatment variation.** Within-county changes in news supply around 2022–26, when nearly all projects occur, are small: 66 announcing counties switch zero status. QCEW's 2022 code change adds measurement noise exactly then.
- **Small outcome counts.** There are 87 dated failures and 72 stalled campuses at 24 months. Confidence intervals on failure effects are wide enough to include meaningful effects of either sign.

- **Grid controls are approximate.** OSM substation voltages are missing for about 22% of substations, and the HIFLD line layer is a 2024 archive; the ≥230 kV line total (300,780 km) looks high relative to published US transmission totals, so line-km is noisy [`output/logs/30_pull_grid_infra.txt`]. Texas delivery-only utilities have no EIA-861 price.
- **Election shares** for 2020/2024 come from an unofficial compilation. Its 2016 cross-check differs from MEDSL by >1 point for about 10% of counties.

# 7. Easy data wins for Daniel

| Done | Item | Why it matters | Effort | How |
|---|---|---|---|---|
| [ ] | **Census API key** (free) | lets scripts use the API instead of bulk files; future ACS/CBP vintages | 5 min | api.census.gov/data/key_signup.html; set `CENSUS_API_KEY` |
| [ ] | **USDA NASS QuickStats key** (free) | avoids the 585 MB bulk file; easy yearly updates | 5 min | quickstats.nass.usda.gov/api |
| [ ] | **MEDSL county returns** (guestbook form) | official 2020/2024 vote shares (current ones are unofficial) | 5 min | Dataverse doi:10.7910/DVN/VOQCHQ → save as `data/raw/medsl/countypres_2000-2024.csv`; `22_pull_elections.py` picks it up (that branch is untested) |
| [ ] | **Media Cloud account + API key** (free, registration) | **project-level coverage counts**: articles mentioning each project, developer or code name in local sources before decisions. This is the treatment the brief actually needs | 1 day to set up queries; scale with project names already in Aterio | mediacloud.org; verify current free-tier limits and local-source coverage before relying on it |
| [ ] | **BU library: NewsBank / Access World News, ProQuest** | historical local-paper full text for the same project-level coverage measure; covers small dailies and weeklies Media Cloud may miss | 1–2 days for a TX/VA pilot of ~100 projects | BU Libraries databases; check text-and-data-mining terms before bulk use |
| [ ] | **Editor & Publisher Yearbook** (BU) | title-level counts by place and year: a better news-supply panel than establishments, and the only route to place-level supply | several days to digitize a few states | BU Libraries (print/online holdings) |
| [ ] | **Email Medill Local News Initiative** | their outlet-level database (6,400+ newspapers, plus public broadcasting, digital and ethnic outlets) is described as "proprietary"; county-year outlet counts would replace the establishment proxy and directly match the 212/1,525 counts | 1 email; response uncertain | localnewsinitiative.northwestern.edu; the 2023 site lists a database manager (John Volk) |
| [ ] | **UNC/Abernathy legacy data** | earlier news-desert database (usnewsdeserts.com, 2018) for 2004–2018 closures | check if still online | usnewsdeserts.com, or ask Medill, where the project moved |
| [ ] | **Data Center Watch** | external opposition outcomes; the free list has 16 projects (10 matched to Aterio; 6 of the matched rows are coded withdrawn, and two of those rows are the same Provident campus) | $250 per quarterly "Insights" report | datacenterwatch.org/shop; do not scrape the map |
| [ ] | **Aterio `PROJECT_PERMIT_URL` → hearing dates (Texas pilot)** | the outcome the project needs | about 2–4 days for TX | filled for 35% of facilities / 21% of campuses (TX 31% of campuses; VA 33%). In TX, 228 URLs on 28 campuses point to agenda/minutes systems (destinyhosted, CivicClerk, Granicus, IQM2, Legistar) where hearing dates and votes can be read [`aterio_links_tx_permit_domains`]. Most other TX permit links are TCEQ/TDLR state filings (dated, but not local hearings) |

| Done | Item | Why it matters | Effort | How |
|---|---|---|---|---|
| [ ] | **Aterio** `DATASHEET_URL` **local-news links** | free proto-coverage measure: 1,800 facilities (25%) cite at least one local-news article; dates and outlets in these links could be harvested | 1 day | `aterio_links_local_news_domains.csv` lists the 253 domains classified as local news, for a human check |
| [ ] | **Aterio events table** (ask vendor) | the data dictionary describes a "Data Centers Events" product with dated Announcement/Construction/Active/Inactive events; it would give stage *histories* instead of a snapshot | email Aterio | not in the current delivery |
| [ ] | **Keep saving Aterio snapshots** | the Feb→Jun comparison already gave the only evidence on secrecy resolution; monthly snapshots turn the stage snapshot into a panel | none | archive each delivery as done in `VINTAGES.csv` |
| [ ] | **CoreLogic via BU** | housing-side outcomes near projects (as planned) | BU access | not evaluated here |
| [ ] | **FCC LMS** (current broadcast) | CDBS is frozen at 2024-01 | manual download in a browser | enterpriseefiling.fcc.gov LMS public database |

# 8. Recommended next steps

Ordered by value per hour:

1. **Texas (+ Virginia) hearing-date pilot from Aterio permit links.**
   - Hand-collect hearing dates and vote outcomes for the 28 TX campuses with agenda/minutes links, plus VA withdrawn/delayed campuses.
   - This tests whether a decision-level outcome can be built at all; without it the intensive margin has no outcome.
2. **Project-level coverage for the same pilot.**
   - Count local articles before each hearing (Media Cloud first; NewsBank via BU for smaller papers).
   - This replaces the establishment proxy with the actual treatment and shows whether coverage varies enough across projects to be useful.
3. **Request the Medill outlet database**, and check whether the UNC 2004–2018 file is still available. Outlet counts by county-year would fix the 816-vs-212 validity gap and make closure events meaningful.
4. **Archive Aterio snapshots monthly** and ask the vendor about the Events table. Stage histories turn "failed/stalled" into dated transitions and let secrecy resolution be studied properly.
5. **Improve grid controls.** Replace OSM substations with a licensed or utility-filed substation inventory with complete voltages, and add interconnection-queue position; re-run with `code/run_all.sh`.

**Observations** (one line each, not pursued):

- Zero-newspaper hosts get larger, more AI- and BTM-heavy projects.
- Newspaper-establishment declines co-move with data-center growth in larger counties.

# 9. Appendix

## 9.1 Variable dictionary

| Variable | Definition | Source / script |
|---|---|---|
| `q_est` | private newspaper-publisher establishments (QCEW annual average; 511110 ≤2021, 513110 ≥2022) | 02b |
| `q_pc10k` | `q_est` per 10,000 residents (PEP) | 04 |
| `q_zero` | 1 if `q_est` = 0 | 04 |
| `q_grp` | 0 / 1 / 2+ establishments | 04 |
| `q_pcq` | within-year quartile of `q_pc10k`; ties share the lowest rank, so all zero counties are in quartile 1 | 04 |
| `q_decl`, `q_1to0`, `q_0to1` | change vs t−1: decline ≥1; ≥1→0; 0→≥1 (CT 2024 set missing) | 04 |

| Variable | Definition | Source / script |
|---|---|---|
| c_est, c_zero | CBP establishments / zero flag, 2005–2016 only | 01b, 04 |
| c_ge3 | CBP ≥3 establishments (consistent all years; CT 2022–23 missing) | 01b |
| *_l1, *_f5 | value in t−1 / t+5 | 40 |
| n_ann, any_ann, mw_ann | campus announcements in county-year; any; summed MW | 40 |
| failed / failed_any | all / any campus buildings Not Approved/Withdrawn or Cancelled | 05 |
| stalled_X | see A5 | 05 |
| undisclosed | campus provider = "Company Not Disclosed" | 05 |
| no_end_user | no building lists END_USER_COMPANIES | 05 |
| detect_lag_m | months from earliest announced date to earliest record creation | 05 |
| mw, mw50, mw_bin | summed SELECTED_POWER_CAPACITY_MW; >50 MW; Aterio bins | 05 |
| ai, btm | any building flagged | 05 |
| log_density, log_income, sh_65plus, sh_ba_plus | ACS 2017–21 | 20, 40 |
| rucc, metro, rural_grp | RUCC 2023 (CT: 2013); metro = RUCC ≤3; groups 1–3 / 4–7 / 8–9 | 21, 40 |
| log_land_value | log NASS 2022 ag land value $/acre | 24, 40 |
| price_ind_cty, price_ind_util | EIA-861 2025 industrial price, county average / project utility | 31 |
| bws_raw_cty, bws_raw | Aqueduct 4.0 baseline water stress, county / project | 32 |
| rep_share_YYYY | Republican share of total presidential vote | 22, 40 |
| n_general_purpose, govs_per10k | CoG 2022 county + municipal + township governments | 23, 40 |
| n_tv_full, n_radio_full | full-power stations licensed to communities in county (CDBS 2024-01) | 11 |

## 9.2 Code map

| Script | Produces |
|---|---|
| 00_profile_aterio.py | aterio_* profile tables; output/logs/00_profile_aterio.txt |
| 00b_download_geo_pop.py | county shapes, CT crosswalk, PEP, Medill PDF |
| 01a/01b_*cbp*.py | CBP raw + cbp_county_year; cbp_* tables |
| 02a/02b_*qcew*.py | QCEW raw + qcew_county_year; qcew_* tables |
| 03_build_population.py | pop_county_year |
| 04_build_news_panel.py | news_county_year; news_* tables |
| 05_build_aterio_campus.py | campus file; a5_* tables |
| 06_overlap_power.py | a3_* tables |
| 10–14 | LoC, FCC, GDELT check, Data Center Watch, Aterio link profile (aterio_links_*, dcw_aterio_match) |
| 20–24 | ACS, RUCC/UIC, elections, Census of Governments, NASS |
| 30–33 | grid (HIFLD lines, OSM substations), EIA-861, Aqueduct, broadband |
| 40_build_analysis_data.py | county controls, county-year panel, campus analysis file |
| 41_measurement.py | a1_*, a2_*, a4_* |
| 42_descriptives.py | b1_*–b4_*, figures b4_*, b5_* |
| 43_extensive.py | c_*, c5_*, c6_*, figure c6_* |
| 44_intensive.py | d_* |
| 45_secrecy.py | e_* |
| 46_placebos.py | f1_*–f3_* |
| 50_build_memo.py | this memo (MD + PDF) |
| run_all.sh | runs everything in order |

## 9.3 Additional regression tables

### Robustness outcomes, intensive margin (coefficients):

| outcome | regressor | (1) bivariate | (2) +project | (3) +county & siting controls | (4) state x year FE | (5) BA x year FE |
|---|---|---|---|---|---|---|
| failed_any | q_pc10k | -0.026 | -0.022 | -0.025 | -0.035 | -0.021 |
| failed_any | q_zero | 0.030 | 0.028 | 0.024 | 0.029 | 0.019 |
| stalled_12 | q_pc10k | -0.057 | -0.046 | -0.111 | -0.075 | -0.078 |
| stalled_12 | q_zero | 0.159 | 0.175 | 0.212 | 0.160 | 0.144 |
| stalled_18 | q_pc10k | 0.017 | 0.027 | -0.039 | -0.019 | 0.014 |
| stalled_18 | q_zero | 0.086 | 0.124 | 0.146 | 0.145 | 0.095 |
| stalled_36 | q_pc10k | -0.233 | -0.246 | -0.343 | -1.172 | -0.489 |
| stalled_36 | q_zero | 0.065 | 0.116 | 0.139 | -0.172 | -0.166 |

### No end user, q_zero:

| term | (1) bivariate | (2) +project | (3) +county controls | (4) state x year FE | (5) BA x year FE |
|---|---|---|---|---|---|
| q_zero | -0.0031 (0.0251) | 0.0119 (0.0252) | -0.0019 (0.0267) | 0.0020 (0.0291) | -0.0215 (0.0375) |
| N | 1059 | 1059 | 1059 | 979 | 724 |

**Leave-one-state-out, all drops:** `output/tables/f3_loso_all.csv` (not printed; 312 rows).

**Long-format files with formulas and clustering for every regression:** `c_all_long.csv`, `d_all_long.csv`, `e_all_long.csv`, `f1_warehousing_placebo.csv`, `f2_placebo_timing.csv`.